

UNIVERSITY OF ECONOMICS AND LAW
FALCUTY OF INFORMATION SYSTEM



Final Project
**Vehicle Insurance Cross-Sell Prediction Among Health
Insurance Customers**

Discipline: Data Analysis with R/Python

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Finally, we wish to thank our parents for their support and encouragement throughout our study!

Commitment

We undertake that this report is completed based on the results of our research and that the results of this report have not been used for any other peer report.

Ho Chi Minh City, November 2025

Group 4

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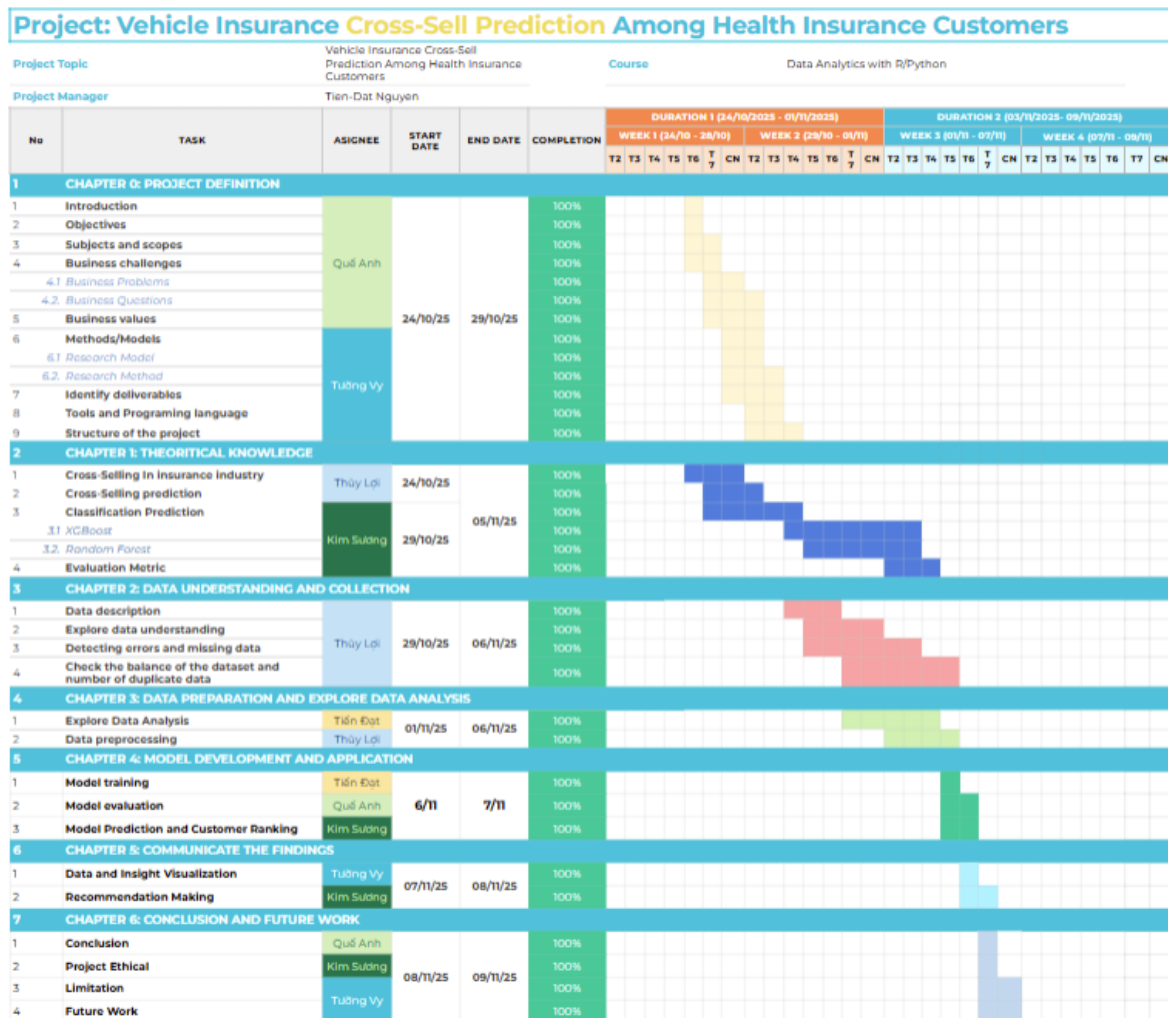
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List of Acronyms

Acronym	Stand For
CAC	Customer Acquisition Cost
CLV	Customer Lifetime Value
CRM	Customer Relationship Management
ROI	Return on Investment
EDA	Exploratory Data Analysis
ROC-AUC	Receiver Operating Characteristic – Area Under the Curve
IQR	Interquartile Range

Gantt chart



Chapter 0: Project Overview

1. Introduction

In today's competitive insurance market, companies are facing increasing pressure from multiple directions: the Customer Acquisition Cost (CAC) continues to rise, competition is intensifying, and customer behavior is becoming more diverse and difficult to predict. As a result, many insurers are shifting their focus from costly customer acquisition to maximizing Customer Lifetime Value (CLV) — turning existing customers into sustainable sources of growth.

One of the most effective strategies for achieving this is cross-selling — offering additional products or services to existing customers. In the insurance industry, individuals who already hold a health insurance policy often have a high potential to purchase complementary products such as vehicle insurance. Cross-selling not only increases revenue per customer but also enhances customer satisfaction and loyalty by providing more comprehensive protection.

To execute cross-selling strategies effectively, insurance companies must overcome a key challenge: accurately identifying customers who are most likely to respond positively to a cross-sell offer. Instead of relying on broad mass marketing campaigns that lead to wasted resources, data analytics and machine learning techniques can help insurers predict which customers are more likely to accept new offers, making marketing more targeted and cost-efficient.

This project leverages the dataset from the health insurance business, which contains 381,109 customer records and 12 variables. Using this dataset, the project aims to build a machine learning model that predicts the probability of a customer, who already has Health Insurance, showing interest in purchasing Vehicle Insurance. This is a classic and practical use case in the insurance sector, directly aligned with data-driven decision-making and customer segmentation strategies.

2. Objectives

The overarching objective of this project is to develop a predictive analytics framework that supports cross-selling strategies in the insurance domain. More specifically, the project aims to identify customers with a high likelihood of accepting a Vehicle Insurance offer, given that they are already holding Health Insurance. This goal is broken down into several clear and structured objectives:

- **Comprehensive data understanding and exploratory analysis**
 - Explore customer demographic and behavioral patterns, including age, gender, vehicle ownership, driving license status, insurance history, region, and sales channels.
 - Assess the relationship between these attributes and the response variable (Response) to uncover underlying patterns.
 - Identify potential data quality issues such as missing values, duplicates, or outliers (especially in Annual_Premium).
- **Develop a robust data preprocessing pipeline**
 - Apply appropriate encoding strategies for categorical and ordinal variables (e.g., Gender, Vehicle_Damage, Vehicle_Age).
 - Handle class imbalance effectively to ensure fair model learning and avoid bias towards the majority class (Response = 0).
 - Implement standardized splitting (Train–Validation–Test) to ensure reliable and reproducible model evaluation.
- **Build and evaluate multiple predictive models**
 - Train a set of machine learning classifiers, including Logistic Regression, Random Forest, XGBoost, and Decision Tree.

- Optimize hyperparameters and apply calibration techniques to generate well-calibrated probability scores.
- Use cross-validation to ensure model stability and generalizability.
- **Adopt business-relevant evaluation metrics**
 - Evaluate models not only with standard metrics such as ROC-AUC but also PR-AUC, which is more appropriate for imbalanced classification problems.
 - Incorporate Precision@K and Lift/Decile charts to assess the marketing impact of selecting top-ranked customers.
 - Identify an optimal threshold that balances precision, recall, and business ROI.
- **Operationalize the prediction output for marketing strategy**
 - Generate a ranked customer list (customer_rank.csv) with predicted probabilities, deciles, and ranking scores.
 - Recommend how to target the top deciles (e.g., top 10%) for campaign execution to maximize conversion rate and minimize cost.
- **Provide actionable business recommendations**
 - Quantify the expected lift in response rate compared to random targeting, providing a business case for model deployment.
 - Propose how the predictive model can be integrated into existing CRM or marketing workflows.
 - Outline opportunities for future model improvements, such as incorporating additional behavioral data or cost-sensitive optimization.
- **Bridge technical modeling with real-world application**

- Ensure that all technical outputs (e.g., model scores, decile analysis) are interpretable for business stakeholders.
- Emphasize transparency, explainability, and scalability to support sustainable, data-driven marketing strategies in the insurance industry.

3. Subjects and Scopes

3.1. Subjects of the study

The primary subject of this study is the application of predictive analytics and machine learning to cross-selling strategies in the insurance industry, with a focus on identifying potential customers who are most likely to purchase Vehicle Insurance among those who already own Health Insurance policies.

The research is based on the Kaggle Health Insurance Cross-Sell Prediction dataset, which consists of 381,109 observations and 12 variables describing customer demographics, vehicle information, historical insurance status, and sales channel identifiers. Each record represents an individual customer and their response to a Vehicle Insurance offer.

The core subject areas include:

- Customer behavior analysis: understanding patterns in demographic and behavioral factors that influence cross-sell decisions.
- Predictive modeling: developing and evaluating machine learning models to predict the Response variable (1 = interested, 0 = not interested).
- Marketing analytics: translating predictive insights into actionable marketing strategies to improve conversion rates and reduce acquisition costs.
- Model interpretability: identifying key factors (e.g., Vehicle Damage history, Vehicle Age, Previously Insured status) that drive customer response behavior.

In addition, the study will emphasize the link between technical modeling and real-world business decision-making, showcasing how predictive models can be used as practical tools for CRM teams, marketing strategists, and insurance product managers.

3.2. Scope of the study

The scope of this research is carefully defined to ensure a balance between technical depth and business relevance.

Data Scope

- Source: the publicly available Kaggle dataset, with no personally identifiable information, ensuring full compliance with privacy and ethical research principles.
- Time dimension: the dataset is static and represents historical customer behavior over a specific campaign period. The study does not include real-time streaming or online learning.
- Variables considered: 12 original features, including demographic (e.g., Gender, Age), vehicle information (Vehicle_Age, Vehicle_Damage), historical insurance status (Previously_Insured), region and channel identifiers, and the target variable Response. No external data sources will be merged at this stage.

Technical Scope

- Machine Learning techniques: Logistic Regression, Decision Tree, Random Forest, and XGBoost classifiers will be used for modeling. These methods are chosen to cover both interpretable linear models and high-performance tree-based ensemble methods.
- Preprocessing: encoding categorical variables, handling class imbalance, scaling numerical variables, and splitting data into training, validation, and test sets.

- Evaluation Metrics: PR-AUC (main), ROC-AUC, Precision@K, Lift/Decile, Confusion Matrix, and business threshold selection based on ROI considerations.
- Model Interpretability: feature importance and SHAP (SHapley Additive exPlanations) analysis will be applied to explain model outputs.

Business Scope

- The model is designed for cross-sell targeting strategy optimization, not for underwriting or pricing decisions.
- The project focuses on ranking customers by their likelihood to accept a Vehicle Insurance offer and providing a top-decile strategy for marketing campaigns.
- The study assumes a campaign-based marketing scenario — i.e., periodic batch scoring of customers — rather than a fully automated, real-time deployment.

3.3. Exclusions and Limitations of Scope

To keep the study focused and feasible within the project timeline, the following aspects are explicitly excluded:

- Integration of external data sources (e.g., transaction logs, claim history, telematics, or social data).
- Real-time scoring and MLOps pipeline development.
- Cost optimization beyond basic lift and ROI estimation.
- Legal and compliance considerations of insurance product delivery.
- Personalization of marketing content (the study focuses on *who to target*, not *what message to deliver*).

These exclusions do not reduce the significance of the study; instead, they establish a clear boundary, allowing the project to deliver a robust, end-to-end predictive modeling framework within academic constraints.

4. Business Challenges

4.1. Business Problems

Customer Targeting Efficiency

One of the most significant challenges in the insurance industry is accurately identifying which customers are most likely to respond positively to cross-sell offers. Unlike acquiring new customers through large-scale marketing, cross-selling requires precise targeting of the right segment at the right time. However, customer behavior is not static. It can fluctuate rapidly depending on income levels, lifestyle changes, risk perception, economic conditions, and previous claim experiences.

This dynamic nature of customer demand makes it difficult for insurance companies to determine which existing policyholders are likely to accept an additional product — such as Vehicle Insurance after purchasing Health Insurance. Inaccurate targeting leads to wasted marketing budgets, lower campaign conversion rates, and customer dissatisfaction caused by irrelevant offers. In a competitive market, not having an effective targeting strategy means losing valuable cross-sell opportunities to competitors.

Furthermore, uncertainty in predicting cross-sell behavior can cause delays in campaign planning, misallocation of resources, and an inability to react to market shifts quickly. If insurers are unable to predict customer interests accurately, they may either target too broad of an audience (leading to inefficiency and high costs) or too narrow (leading to lost potential revenue).

Imbalanced Response Behavior

Another critical challenge is the class imbalance problem in predictive modeling. In real marketing campaigns, only a small percentage of customers respond positively to cross-

sell offers. In the dataset used in this project, the number of positive responses (Response = 1) is significantly lower than negative ones (Response = 0).

This imbalance leads to difficulties in building robust machine learning models. Traditional algorithms tend to bias toward the majority class, resulting in high accuracy but poor precision and recall for the minority class — which is, in fact, the most valuable segment for the business. Addressing this imbalance requires thoughtful use of techniques such as class weighting, resampling, and appropriate evaluation metrics (e.g., PR-AUC, F1-Score, Recall, Precision,...).

Decision-Making Under Uncertainty

Cross-selling decisions often involve limited information and uncertainty. Customers' likelihood of purchasing Vehicle Insurance may depend on many hidden factors that are not fully captured in the dataset (e.g., financial capacity, personal preferences, external economic influences). This lack of perfect information makes it challenging to build predictive systems with high precision.

Moreover, marketers need to decide how many customers to target and which segment of the ranked list will generate the highest return on investment. Without data-driven tools, these decisions rely heavily on intuition, which increases the risk of inefficient budget allocation and missed opportunities.

Operational Integration

Even with accurate predictions, businesses often face challenges in integrating predictive outputs into operational marketing workflows. A predictive model alone is not enough; insurers must develop clear strategies on how to act on the ranked customer list, determine targeting thresholds, and coordinate with sales channels (e.g., telesales, email campaigns, or direct agents). If not aligned properly, the model's impact on real business performance can be limited.

4.2. Business Questions

Based on the business problems identified above, this project focuses on addressing the following key questions:

Question 1: How can the company identify approximately 5,000 potential customers out of 127,038 new health insurance clients who are most likely to purchase Vehicle Insurance, in order to maximize campaign effectiveness and conversion rates?

Question 2: How can marketing teams determine the optimal targeting threshold (e.g., top 5% or top 10% of the ranked list) to balance conversion rate and cost?

Question 3: How can predictive modeling handle class imbalance effectively to improve precision and recall for high-value customers?

Question 4: What are the most influential customer attributes that drive cross-sell interest, and how can these insights guide marketing strategies?

Question 5: How can the predictive model outputs be operationalized to support scalable, data-driven marketing campaigns in real business environments?

5. Business Values

Question 1: How can the company identify approximately 5,000 potential customers out of 127,038 new health insurance clients who are most likely to purchase Vehicle Insurance, in order to maximize campaign effectiveness and conversion rates?

The company currently has a total of 127,038 newly acquired customers, and the marketing objective is to convert approximately 5,000 of them into Vehicle Insurance policyholders. To achieve this goal efficiently, a predictive model is developed to rank customers according to their likelihood of purchasing Vehicle Insurance. This data-driven ranking allows the marketing team to focus resources on customers with the highest probability of conversion instead of using costly and ineffective mass marketing approaches.

By concentrating efforts on the top 10 percent of high-probability customers (around 12,700 individuals), the campaign can achieve an estimated conversion rate of 35 to 40 percent, leading to 4,500 to 5,000 new policies, which aligns with the business target. To provide a safety margin for potential declines in response rates during telesales or email outreach, the marketing team may extend the target range to the top 12 to 15 percent of customers (approximately 15,000 to 19,000 individuals). This strategy can reduce unnecessary contact by about 90 percent, resulting in a substantial improvement in marketing efficiency and return on investment.

Question 2: How can marketing teams determine the optimal targeting threshold (e.g., top 5% or top 10% of the ranked list) to balance conversion rate and cost?

The targeting threshold was determined based on the highest F1-score, which represents the optimal balance between Precision and Recall. This ensures that the model identifies as many truly interested customers as possible while minimizing false positives that could increase campaign cost. At the optimal F1 point (for instance, threshold = 0.42), approximately 12 percent of the 127,038 new health insurance customers are predicted as potential buyers. This subset of around 15,000 customers forms the marketing team's focus group to achieve the target of 5,000 successful Vehicle Insurance conversions.

Question 3: How can predictive modeling handle class imbalance effectively to improve precision and recall for high-value customers?

Cross-sell campaigns in insurance typically involve highly imbalanced data, where the number of positive responders (customers who accept the offer) is much lower than non-responders. Traditional models often struggle with this, resulting in low precision and missed opportunities. By applying methods such as class weighting, oversampling/undersampling, and PR-AUC-based evaluation, the predictive model ensures that the minority class — the most valuable customers — is accurately identified.

This directly improves both precision and recall, meaning fewer wasted contacts and more positive responses captured. From a business standpoint, this means marketing

efforts are focused on customers who are truly interested, lowering the cost per acquisition and increasing overall campaign profitability. In short, addressing class imbalance turns a technical modeling problem into a strategic marketing advantage.

Question 4: What are the most influential customer attributes that drive cross-sell interest, and how can these insights guide marketing strategies?

One of the key advantages of predictive modeling is the ability to explain why certain customers are more likely to respond to offers. By using feature importance analysis and SHAP (SHapley Additive Explanations), the model highlights which variables most influence purchasing decisions — for example, `Previously_Insured`, `Vehicle_Damage`, `Vehicle_Age`, or customer tenure.

These insights allow marketing teams to design targeted segmentation strategies and tailored messaging. For instance, customers with prior insurance experience may respond better to coverage upgrade offers, while those with vehicle damage history may be more interested in protection-focused plans. Moreover, these insights can support product bundling strategies, aligning product offerings with customer needs. As a result, marketing becomes smarter, more personalized, and more effective, improving both conversion and customer satisfaction.

Question 5: How can the predictive model outputs be operationalized to support scalable, data-driven marketing campaigns in real business environments?

Beyond prediction, the model's greatest value lies in its operational integration. Once customer scores and decile rankings are generated, these outputs can be seamlessly integrated into CRM systems, marketing automation platforms, or telesales workflows. This enables marketing teams to execute campaigns at scale, targeting top-priority segments automatically, and tracking performance in real time.

Furthermore, by continuously updating and re-scoring customers, the company can adapt quickly to changes in behavior or market conditions. This transforms traditional campaigns into continuous, data-driven marketing cycles, where targeting decisions are based on evidence rather than guesswork. The result is a scalable, repeatable, and

measurable marketing strategy, supporting sustainable revenue growth and operational efficiency.

6. Methods/Models

6.1. Research method

This study adopts a quantitative research approach based on supervised machine learning techniques to predict customers' likelihood of purchasing Vehicle Insurance given that they already own Health Insurance. The method emphasizes empirical experimentation, statistical evaluation, and business interpretability, ensuring that model outcomes are both technically sound and actionable for marketing decisions.

The research is designed under a data-driven experimental framework, using a large dataset from Kaggle with 381,109 observations and 12 attributes.

The target variable (Response) is binary — “1” indicates interest in Vehicle Insurance, “0” indicates no interest. This allows the use of classification algorithms to estimate purchase propensity and prioritize marketing efforts.

To ensure methodological rigor, all models are trained and evaluated under identical preprocessing and sampling conditions. The research applies four main algorithms:

- Logistic Regression – as a statistical baseline for linear relationships.
- Decision Tree – to interpret hierarchical, rule-based patterns.
- Random Forest – to improve robustness through bagging and feature randomness.
- XGBoost – to enhance predictive performance through gradient boosting and regularization.

6.2. Research model

The research model provides a structured analytical pipeline linking data, machine learning, and business outcomes.

It follows a step-by-step process from raw data preparation to model evaluation and business interpretation.

Research Model

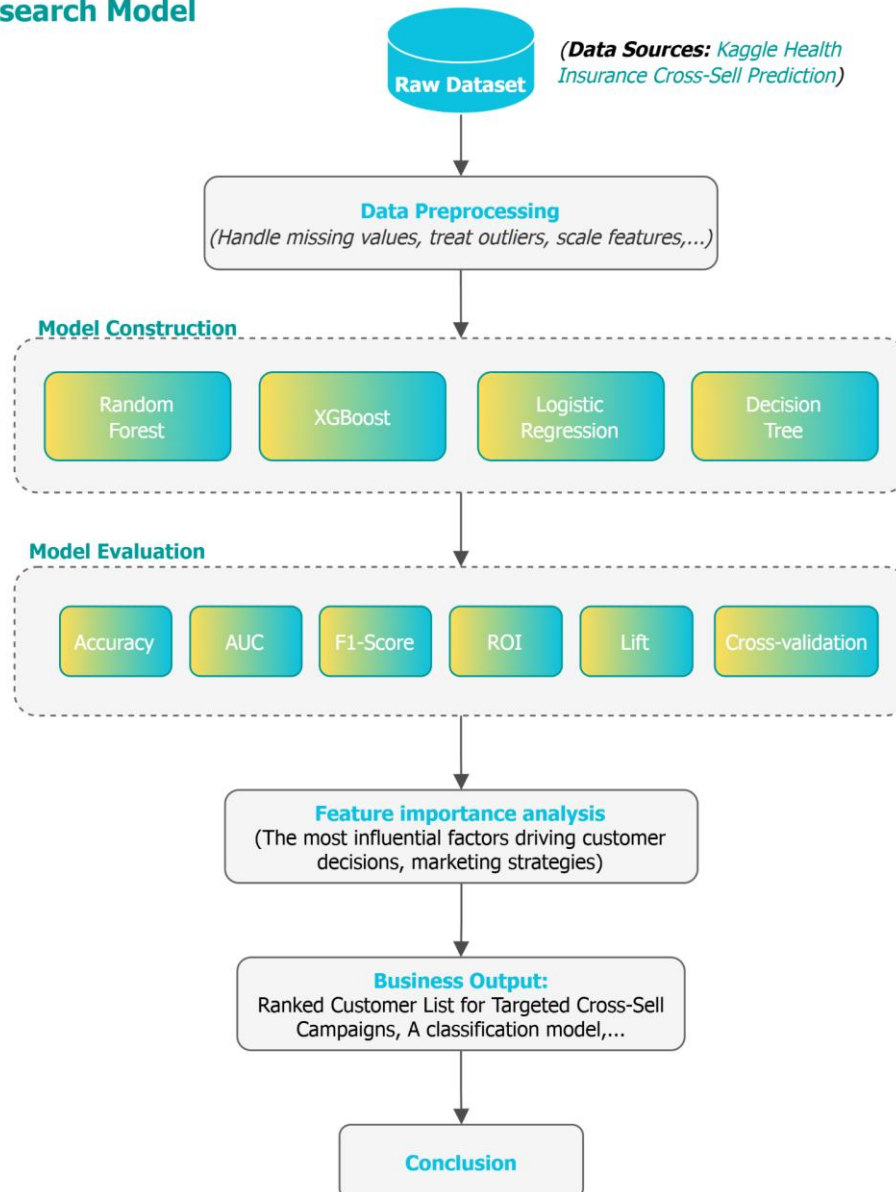


Figure 1. Research Model

Table 1. Description of Research Model

Step	Description
Step 1: Data Understanding	The dataset originates from an insurance company aiming to cross-sell Vehicle Insurance to existing Health Insurance customers. It contains demographic, policy, and behavioral attributes such as <i>Gender</i>, <i>Age</i>, <i>Vehicle_Age</i>, <i>Vehicle_Damage</i>, <i>Annual_Premium</i>, <i>Policy_Sales_Channel</i>, and <i>Previously_Insured</i>.
Step 2: Data Preprocessing	Data is cleaned, encoded, and prepared for modeling: - Handle missing values and outliers (especially in Annual Premium). - Encode categorical variables (Example: Label Encoding for binary features, One-Hot Encoding for nominal variables). - Address class imbalance by resampling or adjusting class weights. - Normalize numerical variables to stabilize gradient-based learning models. - Split data into training ($\approx 80\%$) and testing ($\approx 20\%$) sets.
Step 3: Model Construction	Four models are trained and compared: - Logistic Regression – establishes baseline performance and interpretability. - Decision Tree – identifies hierarchical decision paths.

	- Random Forest – combines multiple trees to reduce variance and overfitting. - XGBoost – optimizes model performance through boosting and regularization. Each model predicts the target variable (Response), estimating the probability of interest in Vehicle Insurance.
Step 4: Model Evaluation	All models are benchmarked using the same validation set and evaluation metrics: - Accuracy, Precision, Recall, F1, AUC-ROC, AUPRC for statistical performance. - Lift Chart and Decile Analysis for business impact measurement (Example: top 10% high-score customers contribute the largest share of conversions). - ROI Analysis to evaluate profitability under different marketing budgets.
Step 5: Feature importance analysis	After evaluating all classification models, the study proceeds with an analysis of feature importance to interpret the drivers of customer purchase behavior. For ensemble algorithms such as Random Forest and XGBoost, feature importance values are computed based on the reduction in impurity or gain across decision splits. The insights obtained from this analysis help the research team and the insurance company understand the key influencing factors behind a customer's decision to purchase Vehicle Insurance.

	Typical high-impact variables include: <table><tr><th>Variables</th><th>Description</th></tr><tr><td>Vehicle_Damage</td><td>A strong indicator of previous claim activity and vehicle risk profile.</td></tr><tr><td>Previously_Insured</td><td>Differentiates loyal customers from first-time buyers, influencing cross-sell potential.</td></tr><tr><td>Vintage</td><td>Measures customer tenure and engagement with the insurer.</td></tr><tr><td>Annual_Premium</td><td>Reflects the customer's financial capacity and price sensitivity.</td></tr><tr><td>Vehicle_Age</td><td>Correlates with insurance renewal likelihood and perceived value.</td></tr></table>	Variables	Description	Vehicle_Damage	A strong indicator of previous claim activity and vehicle risk profile.	Previously_Insured	Differentiates loyal customers from first-time buyers, influencing cross-sell potential.	Vintage	Measures customer tenure and engagement with the insurer.	Annual_Premium	Reflects the customer's financial capacity and price sensitivity.	Vehicle_Age	Correlates with insurance renewal likelihood and perceived value.
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Annual_Premium	Reflects the customer's financial capacity and price sensitivity.												
Vehicle_Age	Correlates with insurance renewal likelihood and perceived value.												
Step 6: Conclusion	The final stage of the research model summarizes the project's outcomes and highlights its contribution to business objectives. The results demonstrate that machine learning algorithms—especially ensemble models like Random Forest and XGBoost—outperform baseline models (Logistic Regression and Decision Tree) in terms of predictive accuracy, interpretability, and marketing ROI. The adoption of these predictive models enables the insurance company to:												

	- Increase cross-sell efficiency by accurately identifying potential buyers. - Enhance customer experience through personalized, data-driven communication. - Promote data-driven culture and scalability.
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7. Identify deliverables

The project provides a set of technical and business deliverables that transform predictive modeling results into practical marketing insights. The main deliverables are as follows:

- **Calibrated Propensity Model:** A supervised classification model capable of predicting the likelihood that existing health insurance customers will purchase vehicle insurance. The model generates a probability score (Response = 1 or 0) for each customer.
- **Ranked Customer Lists per Region/Channel:** Based on predicted probabilities, customers are ranked according to their cross-sell potential. These lists serve as input for marketing and sales departments to prioritize campaign resources efficiently.
- **Business-Oriented Evaluation Report:** A comprehensive performance assessment using both technical metrics (Accuracy, Precision, Recall, F1-score, AUC-ROC) and business metrics (Area Under the Precision-Recall Curve, Lift/Decile, ROI under cost-per-contact). This report ensures that model accuracy aligns with actual business impact.
- **Feature Importance Analysis:** Insights identifying the most influential factors driving customer interest in vehicle insurance. This helps design more targeted and data-driven marketing strategies.

- **Deployment and Pilot Guidelines:** A simple framework for integrating the model into CRM or marketing systems, allowing pilot testing and monitoring performance under real business conditions.

Together, these deliverables ensure that the analytical outputs are explainable, measurable, and directly applicable to business decision-making.

8. Tools and programming language

To implement the predictive models and perform analytical processing, a combination of open-source tools and programming languages was employed.

The selected tools align with the project's requirements for scalability, flexibility, and reproducibility.

Table 2. Description of Tools and Purpose of use

Tool/Language	Purpose of Use
Python	Main programming language for data analysis and model development.
Pandas, NumPy	Data cleaning, preprocessing, and numerical computation.
Scikit-learn	Implementation of classification algorithms (Logistic Regression, Decision Tree, Random Forest) and performance metrics.
XGBoost Library	High-performance gradient boosting framework for model tuning and ensemble learning.
Matplotlib, Seaborn	Data visualization and presentation of feature importance, correlation, and model results.

Google Colab/ Jupyter Notebook	Interactive environment for running code, visualizing outputs, and documenting the analytical workflow.
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9. Structure of the project

The project is structured into six main chapters, each representing a logical stage in the development of the predictive model and the business analysis process. The overall structure ensures a systematic flow — from defining objectives and understanding data, to model construction, evaluation, and communicating actionable insights for business application.

CHAPTER 0: PROJECT DEFINITION

This introductory chapter outlines the foundation of the study, including the project’s background, objectives, scope, and business motivations. It defines the business problem, key challenges, and expected business values of implementing predictive analytics in cross-selling. Additionally, it presents the chosen research methods, identifies key deliverables, specifies the tools and programming languages used, and provides the overall structure of the project.

CHAPTER 1: THEORETICAL KNOWLEDGE

This chapter summarizes the theoretical and conceptual background underpinning the study.

It introduces the concept of cross-selling in the insurance industry, the logic of cross-sell prediction, and the principles of classification modeling. The chapter also discusses key evaluation metrics (Accuracy, Precision, Recall, F1-score, AUC, Lift, and ROI) used to assess model performance in predictive analytics.

CHAPTER 2: DATA UNDERSTANDING AND COLLECTION

This section focuses on understanding the dataset used for modeling — the Kaggle Health Insurance Cross-Sell Prediction dataset containing 381,109 rows and 12 columns.

It provides a detailed data description, explores variable characteristics, and checks for missing values, duplicates, and data imbalance.

The goal of this chapter is to ensure data reliability before moving to model development.

CHAPTER 3: DATA PREPARATION AND EXPLORATORY DATA ANALYSIS

This chapter covers the data preprocessing and exploratory data analysis (EDA) steps. It includes handling missing values, detecting and treating outliers (particularly in *Annual_Premium*), encoding categorical variables, balancing classes, and normalizing features.

The EDA visualizes relationships between variables and identifies patterns that may influence customer purchase decisions, serving as the foundation for model construction.

CHAPTER 4: MODEL DEVELOPMENT AND APPLICATION

This chapter presents the core of the research — the design, training, and evaluation of predictive models. It includes:

- **Model Training:** building classification models (Logistic Regression, Decision Tree, Random Forest, XGBoost).
- **Model Evaluation:** comparing performance across metrics such as Accuracy, Precision, Recall, F1-score, AUC-ROC, and Lift/Decile analysis.
- **Model Prediction and Customer Ranking:** generating probability scores and creating ranked customer lists for targeted cross-sell campaigns.

The chapter demonstrates how the models are benchmarked and how their results translate into measurable business impact.

CHAPTER 5: COMMUNICATE THE FINDINGS

This section focuses on data visualization and insight communication. It summarizes the analytical findings through visual charts and tables, highlighting feature importance and customer behavior patterns. It also provides recommendations for marketing and business strategy, showing how the model can enhance campaign effectiveness and ROI.

CHAPTER 6: CONCLUSION AND FUTURE WORK

The final chapter concludes the project by summarizing its main achievements, discussing limitations, and proposing directions for future improvement. It emphasizes how the predictive model contributes to data-driven decision-making in the insurance sector and how it can be expanded to other cross-sell opportunities such as Life, Travel, or Home Insurance.

Chapter 1: Theoretical Knowledge

1. Cross-Selling in the Insurance Industry

In the contemporary insurance landscape, cross-selling has evolved from a mere sales tactic into a strategic pillar of sustainable growth and customer relationship management. In essence, cross-selling is the practice of promoting additional or complementary products to existing customers, aiming to maximize the economic value derived from each client relationship while simultaneously strengthening customer loyalty (Ngobo, 2017). In insurance, this typically means offering supplementary products, such as vehicle, home, or life insurance, to clients who already hold a primary policy, such as health insurance.

From a theoretical standpoint, cross-selling is grounded in relationship marketing theory, which emphasizes long-term customer engagement and mutual value creation rather than transactional selling. Within this framework, insurers view the customer relationship as a dynamic asset that can be expanded over time through repeated interactions, trust-building, and needs-based product recommendations. As Kumar (2018) asserts, the success of cross-selling depends on understanding not just what customers buy, but why they buy, integrating behavioral insights, demographic analysis, and psychographic segmentation into personalized offerings.

The importance of cross-selling extends beyond firm-level profitability. From a marketing perspective, it provides a means to enhance customer lifetime value (CLV), reduce churn, and optimize acquisition costs. Retaining existing customers and selling them additional policies is significantly more cost-effective than acquiring new ones. McKinsey & Company (2020) highlights that effective cross-selling strategies can raise average customer profitability by 20–30% and increase retention by up to 30%. Furthermore, by broadening the scope of engagement, insurers can gain a more holistic understanding of individual risk profiles, allowing for more precise underwriting and risk management.

In operational terms, cross-selling has become increasingly sophisticated due to the integration of data analytics, CRM platforms, and digital channels. Insurers now rely on advanced segmentation algorithms, purchase history data, and behavioral tracking to identify patterns that predict cross-sell potential. This shift toward data-driven decision-making transforms traditional marketing into predictive and prescriptive analytics, allowing companies to proactively recommend suitable policies based on customer context and life events (PwC, 2021).

From a macroeconomic viewpoint, the role of cross-selling transcends firm boundaries. It contributes to greater financial inclusion by expanding access to multiple forms of protection, especially in emerging markets where insurance penetration remains low. Encouraging customers to diversify their insurance portfolios enhances financial stability at both household and national levels, reducing the socioeconomic impact of unforeseen events such as accidents, illnesses, or natural disasters (OECD, 2022). Cross-selling thus aligns with global financial resilience objectives and supports the long-term development of the insurance ecosystem.

In addition, cross-selling facilitates risk pooling and diversification, which are essential principles of actuarial science. By promoting multi-policy ownership, insurers can balance their risk exposure across various product lines, leading to more predictable revenue streams and improved solvency ratios. This, in turn, strengthens systemic stability within the insurance sector.

Ultimately, cross-selling in the insurance industry represents an intersection of marketing strategy, behavioral economics, and data science. Its successful implementation depends not only on advanced analytical tools but also on customer-centric culture, ethical transparency, and compliance with data protection regulations. When executed effectively, cross-selling not only increases organizational profitability but also enhances the social value of insurance by ensuring broader, deeper, and more equitable protection coverage.

2. Cross-Selling Prediction

The evolution of cross-selling prediction marks a significant leap in how insurers leverage technology to anticipate customer needs and personalize product offerings. At its core, cross-selling prediction refers to the application of statistical modeling, machine learning (ML), and artificial intelligence (AI) to forecast the likelihood that a given customer will purchase an additional insurance product. This predictive capability transforms conventional marketing from reactive communication into proactive engagement, where insurers can identify potential buyers before they even express interest (Lemmens & Gupta, 2020).

The foundation of cross-selling prediction lies in the analysis of historical and behavioral data. By integrating multiple dimensions, such as demographics, policy tenure, claim history, premium level, and engagement frequency, insurers can construct predictive models capable of distinguishing between customers who are likely and unlikely to respond to new product offers. Classical models such as logistic regression provide interpretable insights into the factors influencing purchase decisions, while modern techniques such as Random Forests, Gradient Boosting Machines (GBM), and Neural Networks capture non-linear interactions and higher-order dependencies (Verbeke, Martens, & Baesens, 2012).

From a managerial perspective, the implementation of predictive cross-selling yields measurable marketing advantages. It enhances targeting precision, thereby improving campaign efficiency and reducing wasted expenditure on uninterested segments. According to Kumar and Petersen (2012), predictive customer analytics enables firms to allocate marketing budgets more effectively by focusing on high-propensity customers, resulting in superior return on marketing investment (ROMI). Additionally, prediction models allow for the automation of recommendation systems, where real-time analytics personalize offers based on customer behavior and engagement across digital platforms.

The implications of predictive modeling extend beyond marketing optimization to include strategic decision-making and risk management. For instance, identifying

which customers are more inclined to purchase vehicle insurance in addition to health insurance provides insurers with actionable insights for portfolio diversification and risk balancing. It also allows companies to anticipate potential claims behavior, enabling more precise pricing and product bundling strategies.

Furthermore, the integration of ethical AI and explainable machine learning (XAI) frameworks has become essential in ensuring fairness and regulatory compliance. Predictive cross-selling must align with legal standards such as the EU General Data Protection Regulation (GDPR) to ensure transparency and accountability in automated decision-making (European Commission, 2020). This emphasis on responsible AI promotes trust and protects consumers from discriminatory or manipulative targeting practices—an increasingly important concern as personalization technologies grow more powerful.

From an organizational development standpoint, cross-selling prediction systems represent a convergence of marketing intelligence, data engineering, and actuarial analytics. Insurers that successfully operationalize predictive systems benefit from a feedback loop of continuous learning, where each marketing campaign refines the predictive model through new behavioral data. Over time, this leads to increasingly accurate forecasts, higher conversion rates, and more efficient cross-sell strategies.

On a macro level, predictive modeling also supports broader financial stability by improving resource allocation across the industry. By predicting customer needs more accurately, insurers can develop tailored products that fill coverage gaps, promote responsible insurance behavior, and ultimately enhance societal resilience to financial risks.

In conclusion, predictive cross-selling represents an advanced fusion of marketing science and artificial intelligence, empowering insurers to make informed, data-driven decisions while maintaining ethical integrity and regulatory compliance. It enables firms to transition from intuition-based selling to evidence-based marketing, achieving both business efficiency and customer satisfaction. This paradigm shift underscores the

growing role of analytics and AI in shaping the future of the insurance industry, where competitive advantage increasingly depends on the intelligent use of data.

3. Classification Model

In the context of marketing, binary classification is frequently used to predict whether a customer will buy (or not buy) a product, subscribe (or not) to a service, or respond (or not) to a campaign — the typical “will buy / will not buy” scenario. From a real-world perspective, classification plays a critical role in cross-selling, churn prediction, lead scoring and campaign targeting.

3.1. Random forest

Random Forest is an ensemble learning method that constructs a multitude of decision trees using bootstrap-aggregated samples and random subsets of features, and combines their outputs.

Mechanism:

- Each tree in the random forest is trained on a bootstrap sample drawn from the original training data.
- At each split within a tree, a randomly selected subset of the available features is considered, which introduces feature-level randomness and reduces correlation among trees.
- At prediction time, for classification tasks, each tree “votes” for a class and the class receiving the majority of votes is output; for regression tasks, the average of the trees’ predictions is taken.

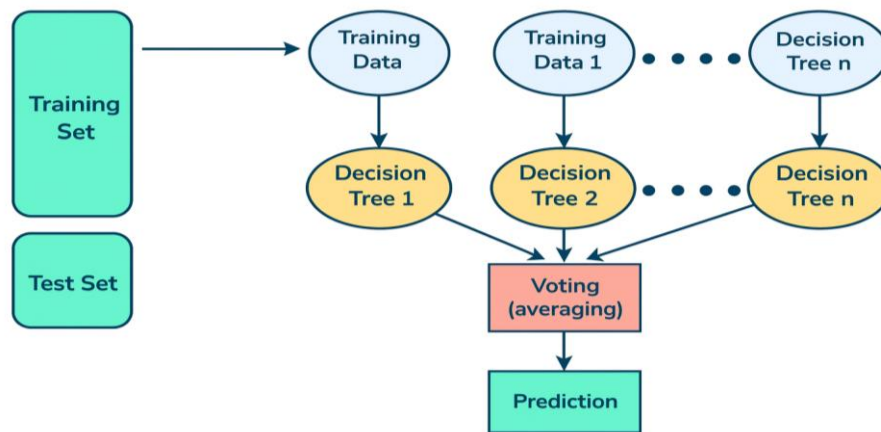


Figure 2. Random Forest Algorithm Working

In a marketing context such as cross-selling, lead scoring, or churn prediction, the Random Forest model provides strategic value by enabling data-driven decision-making. Through its feature-importance analysis, marketers can identify the key factors that influence customer behavior, such as purchase history, demographics, or engagement level, and refine targeting strategies accordingly. Moreover, by generating probability-based customer scores, Random Forest supports the development of effective scoring frameworks that rank customers by their likelihood to respond. This allows firms to allocate marketing budgets more efficiently toward high-potential segments, thereby optimizing cost per acquisition (CPA) and maximizing return on investment (ROI).

Table 3. Advantages and disadvantages of Random Forest

Advantages	- Combines many trees to reduce overfitting and improve accuracy. - Works well with large, complex, or incomplete data and needs little preprocessing. - Shows which features are most important for making predictions.
Disadvantages	- Needs more computing power and time because it trains many trees. - Harder to explain than a single decision tree since the logic is spread across many trees.

3.2. XGBoost

XGBoost is an advanced implementation of the gradient boosting algorithm in which sequential decision trees are built where each new tree corrects the errors of the preceding ensemble.

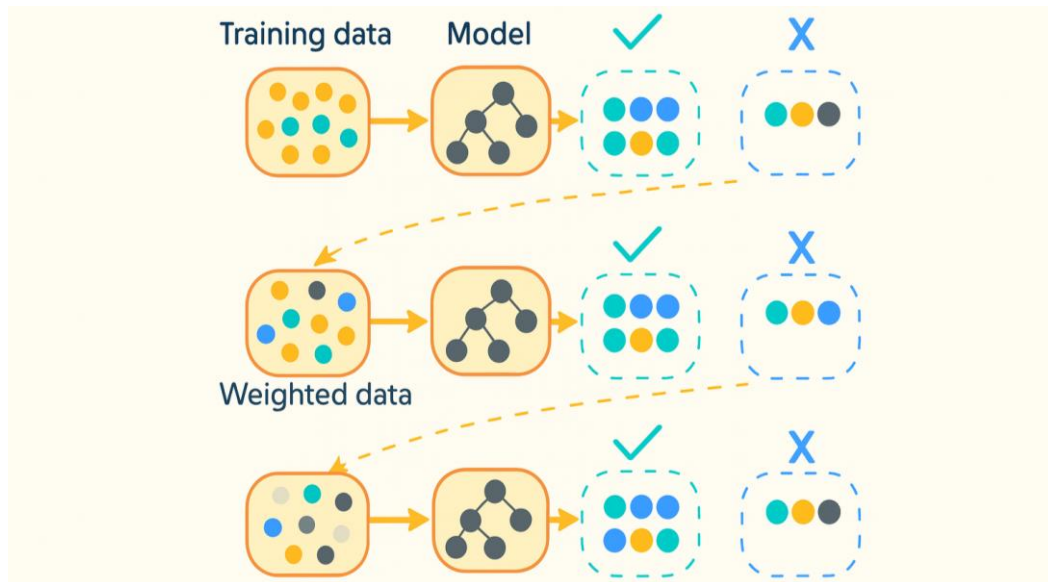


Figure 3. XGBoost core mechanism

Mechanism:

- The algorithm applies gradient descent to minimise a defined loss function by adding trees in an additive manner.
- It incorporates regularisation terms (L1 and L2) to penalise model complexity and reduce overfitting.
- It supports efficient handling of missing values, parallel computation, sparsity-aware algorithms, and scaling to large datasets.

Table 4. Advantages and disadvantages of XGBoost

Advantages	- High performance with large and complex datasets; often delivers superior predictive accuracy. - Better handling of imbalanced data via boosting focus on difficult cases.
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	- Flexibility for ranking problems thanks to its ability to output finely ranked scores.
Disadvantages	- More complex to tune: many hyper-parameters must be optimised for best performance. Less interpretable than simpler models which can complicate understanding and stakeholder communication.

In a marketing context, XGBoost supports the generation of propensity scores for customers. By ranking customers by score, a business can identify the “Top K” high-potential segment and allocate marketing budget to those leads to optimise campaign spend and maximise return on investment.

4. Evaluation Metric

Evaluation metrics play a critical role in assessing the performance of classification models, ensuring that predictions are both statistically reliable and practically meaningful. In imbalanced datasets, common in real-world business contexts such as insurance cross-selling, accuracy alone can be misleading, as a model predicting the majority class may still achieve a high score without truly identifying valuable customers. Therefore, model evaluation must extend beyond accuracy to consider more informative measures.

Evaluation metrics can generally be divided into two main categories:

- **Statistical metrics**, which focus on the model’s predictive ability (e.g., Precision, Recall, F1-score, AUC).
- **Business-oriented metrics**, which measure the model’s contribution to decision-making and marketing outcomes (e.g., Precision@K, Lift, ROI).

Table 5. Statistical Metrics for Classification Evaluation

Metric	Formula	Meaning	Marketing Interpretation
Accuracy	$(TP+TN) / (TP+FP+FN+TN)$	Overall correctness of predictions.	General performance indicator, but unreliable with imbalanced data.
Precision	$TP/(TP+FP)$	Proportion of predicted positives that are true positives.	Measures how correct the “will buy” predictions are.
Recall	$TP/(TP+FN)$	Fraction of actual positives that are correctly predicted.	Shows how many actual buyers are correctly identified.
F1-score	$2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$	Harmonic mean of precision and recall.	Balances precision and recall for cost-effective targeting.
ROC-AUC / PR-AUC	—	Area under ROC or Precision-Recall curve	Overall discrimination; PR-AUC preferred for imbalanced data

Table 6. Business-Oriented Metrics for Cross-Selling Prediction

Metric	Formula / Logic	Meaning	Marketing Interpretation
Precision@K	$TP@K/K$	Percentage of actual buyers within the	Measures how well the model selects the

		top-K customers ranked by the model.	top 5–10% most likely buyers.
Lift & Decile Chart	Average Response Rate / Response Rate in Segment	Compares model performance with random selection. Lift > 1 means better than random.	Shows how much better the model performs than random targeting.
Profit / ROI-Based Metric	$(TP \times \text{Gain}) - (FP \times \text{Cost})$	Estimates financial impact of predictions.	Indicates the net financial gain after marketing costs.
Model Calibration / Reliability Curve	—	Measures how well predicted probabilities match actual outcomes.	Checks if predicted probabilities match actual buying behaviour.

In insurance cross-selling, where the proportion of customers who purchase additional policies is typically low, ranking-based indicators such as Precision@K, Lift, and PR-AUC are far more insightful than overall Accuracy. These metrics help identify a small but profitable subset of customers, enabling insurers to allocate marketing budgets effectively and maximize return on investment.

Chapter 2: Data understanding and collection

1. Data Description

In the insurance industry, accurately identifying customers who are likely to purchase additional insurance products is crucial for revenue optimization and strategic customer targeting. Similar to the concept of backorder prediction in supply chain management—where the objective is to detect products at risk of going out of stock to prevent operational disruption—this study aims to uncover behavioral patterns among existing health insurance customers to predict their interest in purchasing vehicle insurance. Instead of predicting stock shortages, our predictive task focuses on customer conversion likelihood, which directly influences cross-selling effectiveness and long-term customer value.

The dataset used in this study contains historical records of policyholders who currently hold health insurance plans with the company. It includes demographic information, past insurance behavior, vehicle-related indicators, and policy characteristics. The ultimate goal is to build a predictive model that can determine whether a customer will be interested in purchasing Vehicle Insurance, enabling the company to proactively design tailored marketing strategies and minimize resource inefficiency.

2. Explore data understanding

For the effective application of machine learning techniques, it is essential to gain a comprehensive understanding of the characteristics and structure of the dataset. In statistical and predictive modeling contexts, variables are generally categorized into two main types: categorical variables and numerical variables. This classification not only determines the preprocessing techniques to be applied but also affects model selection and evaluation strategy.

The dataset employed in this study represents historical records of individual customers who currently possess health insurance policies. Each row characterizes a single customer, capturing demographic attributes, vehicle-related indicators, policy engagement history, and behavioral features that may influence their likelihood of

purchasing vehicle insurance. The dataset contains a total of 13 variables, including 12 input features and one target variable, labeled Response, which indicates whether the customer is interested in purchasing vehicle insurance (1 = Yes, 0 = No).

After conducting an exploratory analysis, we identified 5 categorical variables and 7 numerical variables, as shown below:

Categorical Variables: Gender, Driving_License, Region_Code, Previously_Insured, Vehicle_Age, Vehicle_Damage, Policy_Sales_Channel, Response (Target Variable)

Numerical Variables: Age, Annual_Premium, Vintage

Table 7 provides an overview of the features, including their description and data type classification.

Table 7. An Overview of Variables in the Dataset

Feature Name	Description	Type
id	Unique identifier for each customer	Identifier
Gender	Gender of the customer	Categorical
Age	Age of the customer in years	Numerical
Driving_License	1 = Holds a valid driving license; 0 = Does not hold a license	Categorical
Region_Code	Encoded region of the customer's residence	Categorical
Previously_Insured	1 = Already has vehicle insurance; 0 = Does not have vehicle insurance	Categorical
Vehicle_Age	Age category of the customer's vehicle (e.g., <1 year, 1–2 years, >2)	Categorical
Vehicle_Damage	1 = Vehicle damaged previously; 0 = No past damage	Categorical

Annual_Premium	Yearly premium amount to be paid by the customer	Numerical
Policy_Sales_Channel	Encoded marketing/distribution channel used to acquire the customer	Categorical
Vintage	Number of days the customer has been associated with the company	Numerical
Response (Target)	1 = Interested in vehicle insurance; 0 = Not interested	Categorical

Understanding the nature of each variable is essential to determine the appropriate feature encoding technique, identify data quality issues, and ensure optimal model performance. In the following sections, we will further explore the statistical distribution of numerical features, analyze category frequencies, and detect any missing or imbalanced data that may impact the predictive modeling process.

3. Detecting errors and missing data

Ensuring data completeness is an essential step prior to model development, as missing or inconsistent values can negatively impact the performance, reliability, and generalizability of the predictive model. To assess the quality of the dataset, we conducted a preliminary audit focusing on missing values across all variables. The results indicate that the dataset is fully complete, with no missing values detected in any of the features or the target variable.

Table 8. Check for faulty and missing values in data set

Column	Null
id	0
Gender	0
Age	0

Driving_License	0
Region_Code	0
Previously_Insured	0
Vehicle_Age	0
Vehicle_Damage	0
Annual_Premium	0
Policy_Sales_Channel	0
Vintage	0
Response	0

The absence of missing values suggests that the dataset exhibits a high degree of completeness, thereby eliminating the need for imputation or data removal techniques at this stage. This characteristic enhances the credibility of further exploratory data analysis (EDA) and modeling steps, as it ensures that all variables can be used directly without concerns of bias or information loss due to missingness.

Additionally, no anomalies or inconsistent data types were detected, indicating that the dataset is clean and well-structured for subsequent preprocessing and feature engineering.

4. Check the balance of the dataset and number of duplicate data

To ensure the quality and reliability of the dataset, we conducted an analysis to detect any duplicate entries and to evaluate the balance of the target variable. The results indicate that there are no duplicate records based on the unique customer identifier (id), confirming that the dataset is free from redundancy and does not require any deduplication processing.

```

=== Dataset Balance Check ===
      Count  Percentage (%)
Response
0      334399      87.743664
1       46710      12.256336

=== Number of Duplicates ===
Duplicate rows based on ID column: 0

```

Figure 4. Balance and Duplicate Check

However, upon examining the distribution of the target variable (Response), it is evident that the dataset is significantly imbalanced. Specifically, the proportion of customers who are not interested in purchasing vehicle insurance (class 0) accounts for 87.74% of the data, while those who are interested (class 1) represent only 12.26%. This corresponds to an approximate ratio of 1:7, indicating that the negative class is heavily dominant.

Such imbalance poses a challenge for machine learning models, as they may become biased toward the majority class and fail to correctly identify the minority class, which is the primary area of interest in this predictive task. If left unaddressed, this imbalance can lead to poor model generalization, reduced predictive accuracy for the positive class, and ultimately result in suboptimal business decisions.

Therefore, to develop a more robust and reliable classification model, it is necessary to apply appropriate techniques to handle class imbalance, such as oversampling, undersampling, or the use of class-weighted algorithms in subsequent modeling stages.

Chapter 3: Data preparation and explore data analysis

1. Explore Data Analysis

EDA helps you get an overview of the data, key features, correlations between variables, assumptions about the data, and what to do to prepare for analysis or model building.

To better understand the variables in the data set, we will analyze each variable and draw a statistical graph.

1.1. Distribution of Categorical Features

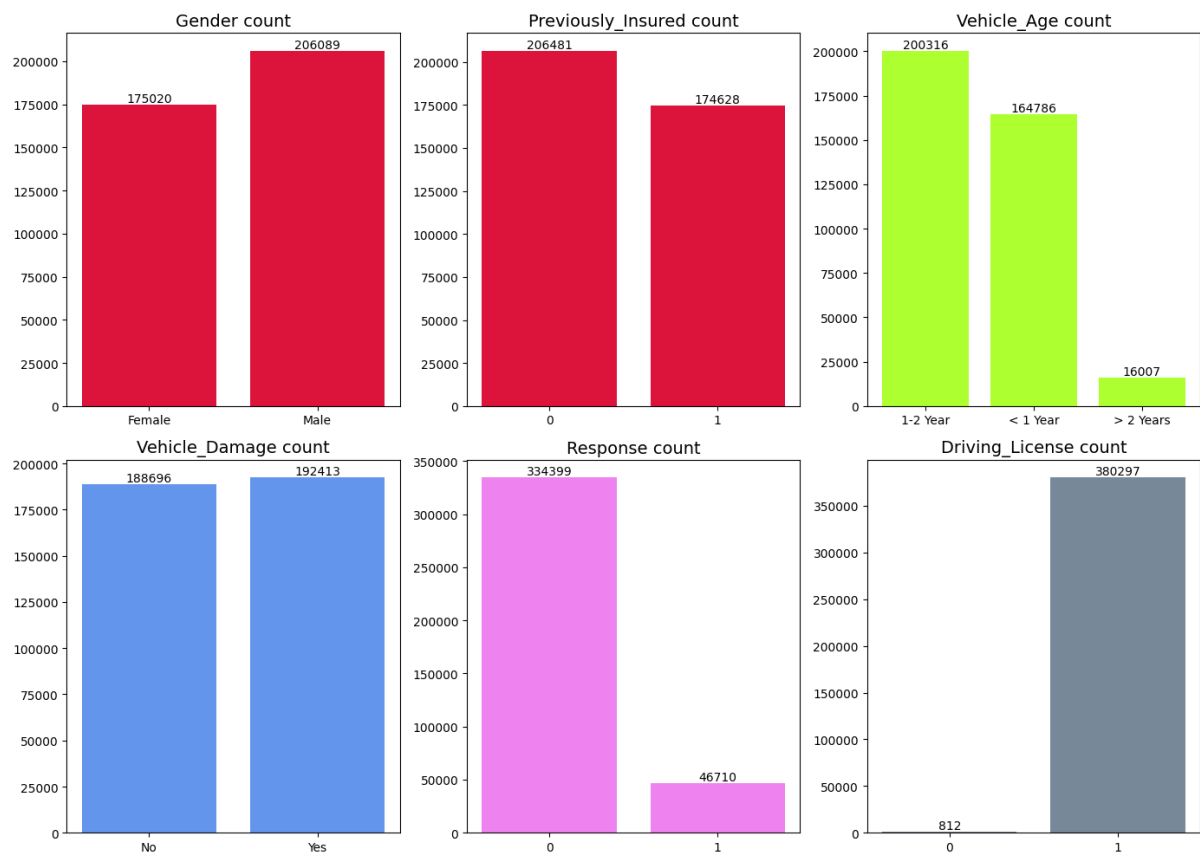


Figure 5. Distribution of Categorical Features

The categorical feature distribution provides an overview of the composition and balance of various attributes within the dataset.

Starting with Gender, the dataset contains a slightly higher proportion of males (206,089) compared to females (175,020), indicating a modest gender imbalance. The

Previously_Insured feature shows that the majority of customers (206,481) were not previously insured, while 174,628 had prior insurance coverage.

For Vehicle_Age, most vehicles fall within the 1–2 year range (200,316), followed by those less than 1 year old (164,786). A relatively small portion (16,007) of vehicles are older than 2 years, suggesting that the dataset is dominated by newer vehicles.

The Vehicle_Damage variable displays a nearly balanced distribution between vehicles that were previously damaged (192,413) and those that were not (188,696). In contrast, the Response variable, which represents whether a customer is interested in a new insurance policy, is highly imbalanced — only 46,710 customers responded positively (1), while 334,399 did not (0). This imbalance should be considered in subsequent modeling steps.

Finally, the Driving_License feature shows that almost all customers (380,297) possess a valid driving license, with only a negligible number (812) not holding one.

Overall, the categorical features reveal a few imbalances (notably in the response and driving license variables) and provide important insights into customer demographics and vehicle characteristics, which will be useful for further analysis and model building.

1.2 Distribution of Numerical Features

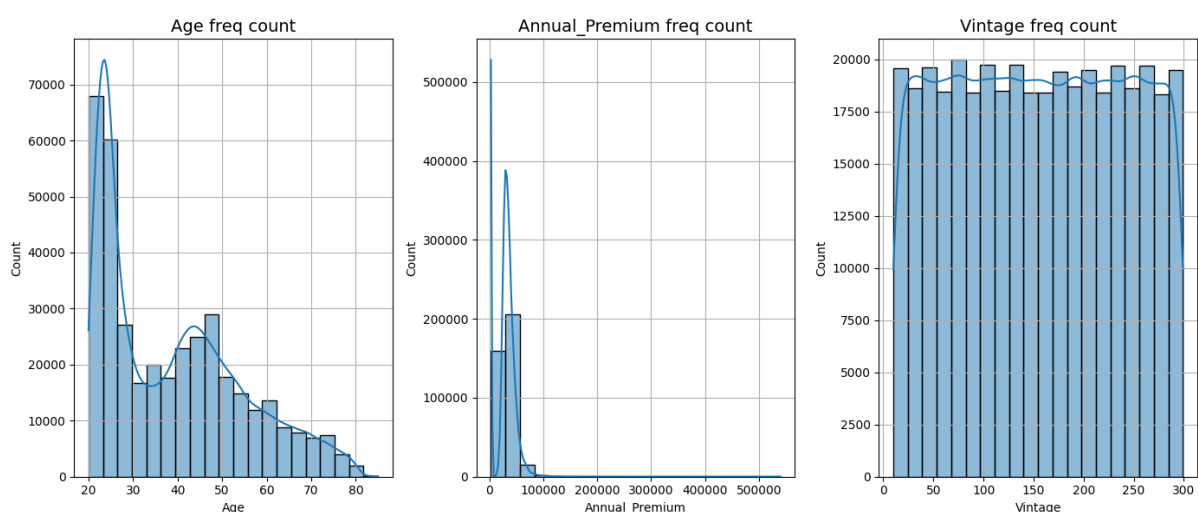


Figure 6. Distribution of Numerical Features

The figure illustrates the frequency distributions of the three numerical features — Age, Annual_Premium, and Vintage — providing insights into their spread and patterns across the dataset.

The Age distribution is right-skewed, with a large concentration of customers between 20 and 30 years old. The frequency gradually declines with age, though a secondary peak appears around the 45–50 age range, indicating another significant customer segment among middle-aged individuals.

The Annual_Premium feature also shows a highly right-skewed distribution, where most customers have annual premiums below 50,000. A few extreme outliers with very high premium values are observed, suggesting the presence of high-value policies that may require normalization or transformation during preprocessing.

In contrast, the Vintage variable, which represents the number of days a customer has been associated with the company, displays a nearly uniform distribution. This suggests that customers across different tenures are evenly represented in the dataset, with no specific time-related bias.

Overall, the numerical feature distributions reveal right-skewed patterns for Age and Annual_Premium, while Vintage remains uniformly spread — valuable information for understanding customer demographics and preparing data for modeling.

1.3. Previous Insurance Status and Response

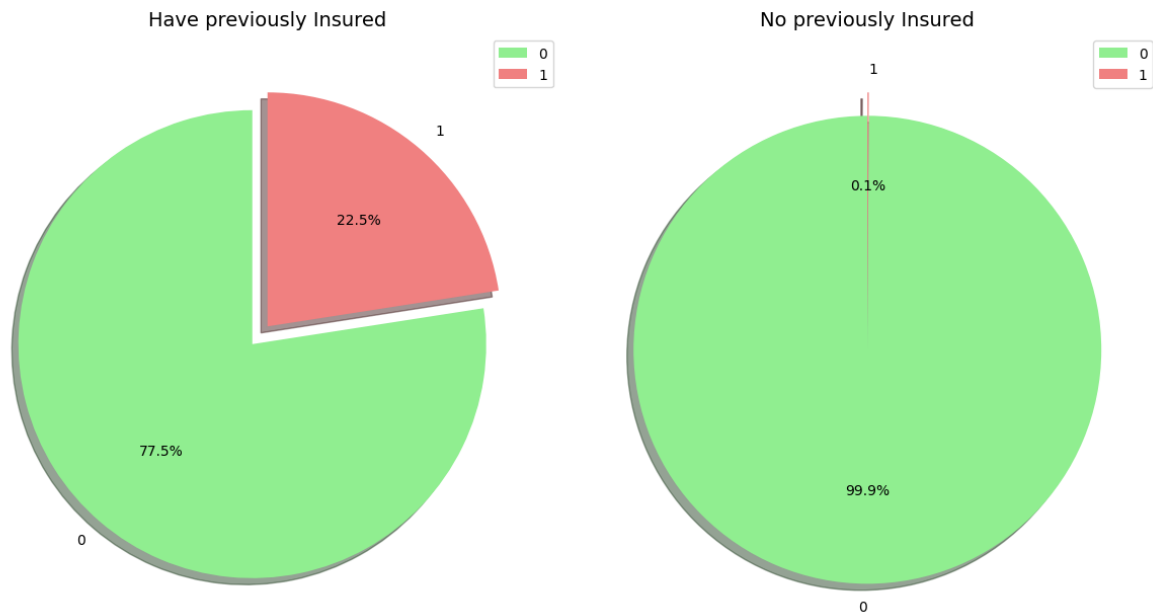


Figure 7. Previous Insurance Status and Response

The figure illustrates the response distribution based on customers' previous insurance status, divided into two groups: those who have previously been insured and those who have not.

In the “Have previously insured” group (left pie chart), 77.5% of customers did not respond positively (label 0), while 22.5% showed interest in purchasing a new insurance policy (label 1). This indicates that previously insured customers are relatively more likely to respond positively to the offer.

Conversely, in the “No previously insured” group (right pie chart), the response distribution is extremely imbalanced — 99.9% did not respond positively, and only 0.1% showed interest. This suggests that customers without prior insurance experience are significantly less likely to engage with the new insurance offer.

Overall, the comparison highlights a strong relationship between previous insurance status and response behavior. Customers with prior insurance coverage tend to exhibit much higher interest in renewing or purchasing additional insurance compared to those who have never been insured.

1.4. Annual Premium by Vehicle Age and Response

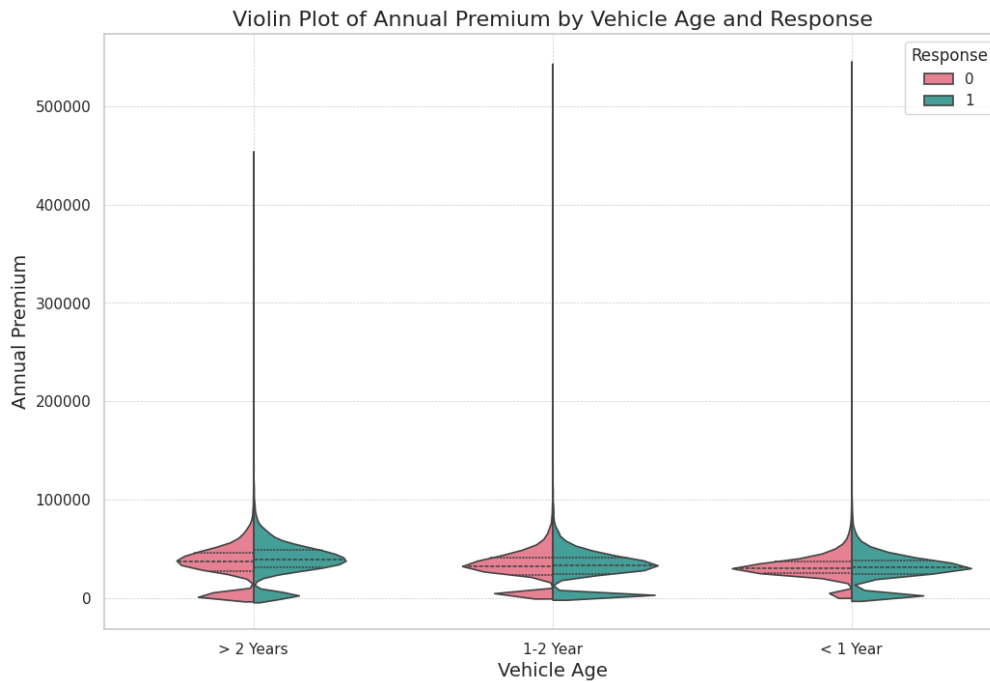


Figure 8. Annual Premium by Vehicle Age and Response

The figure presents a violin plot showing the distribution of Annual Premium across different Vehicle Age categories, separated by customer Response (0 = not interested, 1 = interested).

Across all three vehicle age groups (< 1 Year, 1–2 Years, and > 2 Years), the Annual Premium distributions exhibit similar shapes, indicating that the premium amount does not vary significantly with vehicle age. The majority of premiums are concentrated between approximately 20,000 and 50,000, with a few high-value outliers exceeding 400,000, suggesting the presence of extreme premium values.

When comparing responses, customers who responded positively (Response = 1) tend to have a slightly higher Annual Premium median and broader spread than non-respondents, particularly in the 1–2 Years and > 2 Years categories. This pattern may imply that customers paying higher premiums or owning slightly older vehicles are marginally more inclined to respond positively to the insurance offer.

Overall, the violin plots reveal that while Annual Premium distributions are generally consistent across vehicle ages, there is a modest indication that customers with higher premiums exhibit a greater likelihood of engagement.

1.5. Vehicle Age and Vehicle Damage

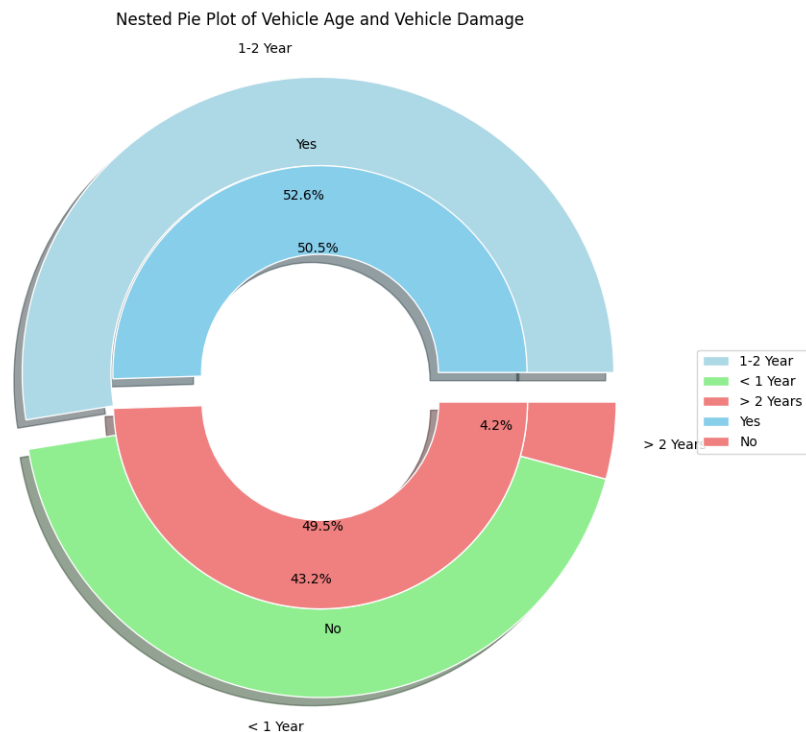


Figure 9. Vehicle age and Vehicle damage

The nested pie chart illustrates the relationship between Vehicle Age and Vehicle Damage, providing a visual comparison of how the likelihood of vehicle damage varies across different age categories.

In the outer ring, the majority of vehicles fall within the 1–2 year (50.5%) and < 1 year (43.2%) categories, while only 4.2% of vehicles are older than 2 years, showing that most vehicles in the dataset are relatively new.

The inner ring shows the proportion of vehicles that have experienced damage (Yes) versus those that have not (No) within each vehicle age group. Notably, 52.6% of the total vehicles have recorded previous damage, while 49.5% have not. The near balance

between the two suggests a fairly even distribution of damaged and undamaged vehicles, regardless of vehicle age.

However, visually, vehicles aged 1–2 years appear slightly more associated with previous damage compared to the newer group (< 1 year). This trend may indicate that as vehicles age slightly, their exposure to damage incidents increases.

Overall, the nested pie plot reveals that most vehicles are relatively new, with a modestly higher damage occurrence among those in the 1–2 year age range. This insight can be useful in understanding how vehicle age correlates with damage likelihood, potentially influencing insurance risk assessment.

1.6. Region Code

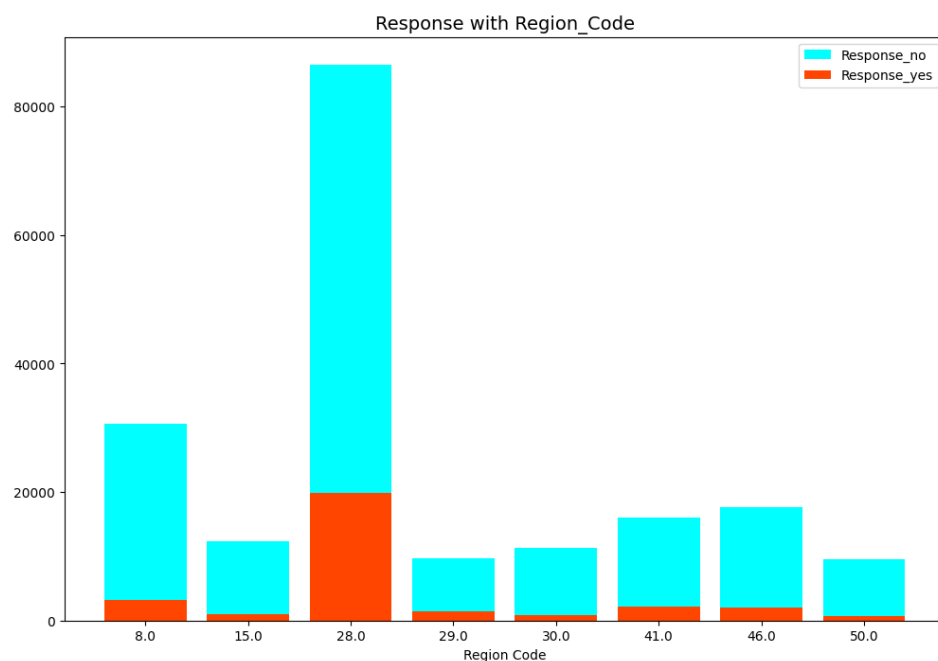


Figure 10. Region Code EDA

The bar chart visualizes the relationship between Region_Code and Response, showing how customer interest (response) varies by region. The turquoise bars represent customers who did not respond positively (Response = 0), while the orange bars indicate those who responded positively (Response = 1).

From the visualization, it is evident that Region Code 28 has the largest customer base overall, with a notably higher number of both responses and non-responses compared

to other regions. This suggests that this region may represent a densely populated or highly targeted area.

Other regions, such as 8, 41, and 46, also show moderate customer volumes but with fewer positive responses. In contrast, regions like 15, 29, 30, and 50 exhibit relatively smaller customer counts, indicating lower participation or fewer policyholders in those areas.

Overall, the response pattern appears consistent across regions—most customers did not respond positively, highlighting a strong class imbalance. However, the concentration of responses in Region 28 suggests that specific regions may have a higher potential for customer engagement and could warrant targeted marketing or outreach strategies.

1.7. Region Code Distribution

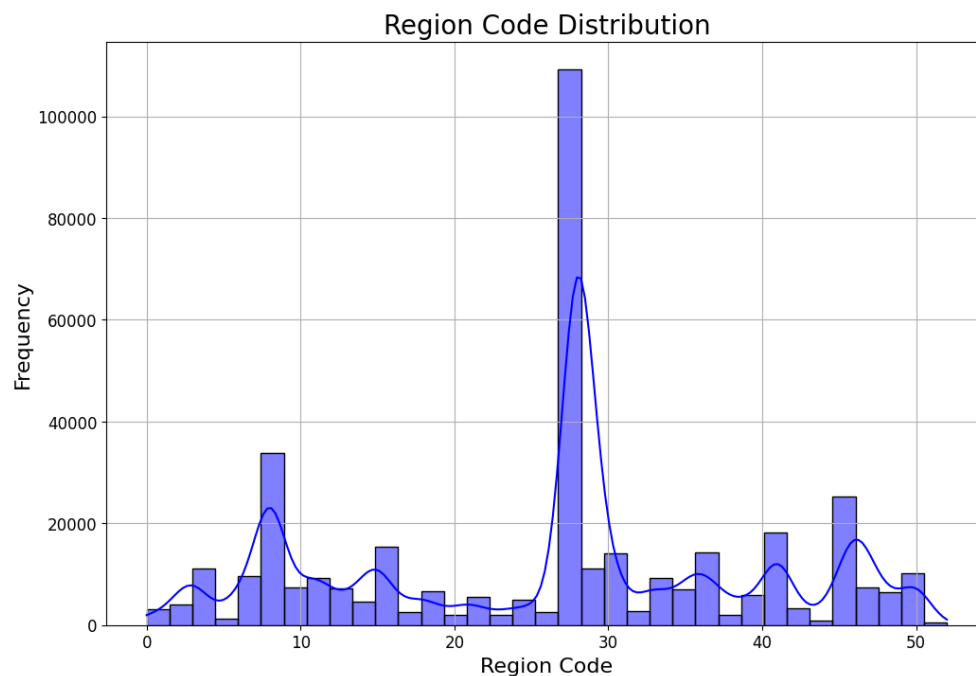


Figure 11. Region Code Distribution

The histogram illustrates the distribution of customers across different Region Codes, highlighting how the dataset is spread geographically.

The chart shows that the distribution is uneven, with some region codes having a much higher frequency than others. In particular, Region Code 28 stands out prominently,

representing the highest customer count, significantly surpassing all other regions. Several other regions, such as 8, 41, and 46, also show moderate concentrations of customers, while the rest have relatively low frequencies.

This pattern suggests that customer data is heavily concentrated in a few regions, possibly reflecting population density, business focus, or market segmentation. Understanding this uneven regional distribution is important for assessing potential regional bias and identifying areas for targeted marketing or customer acquisition strategies.

1.8. Correlation Heatmap

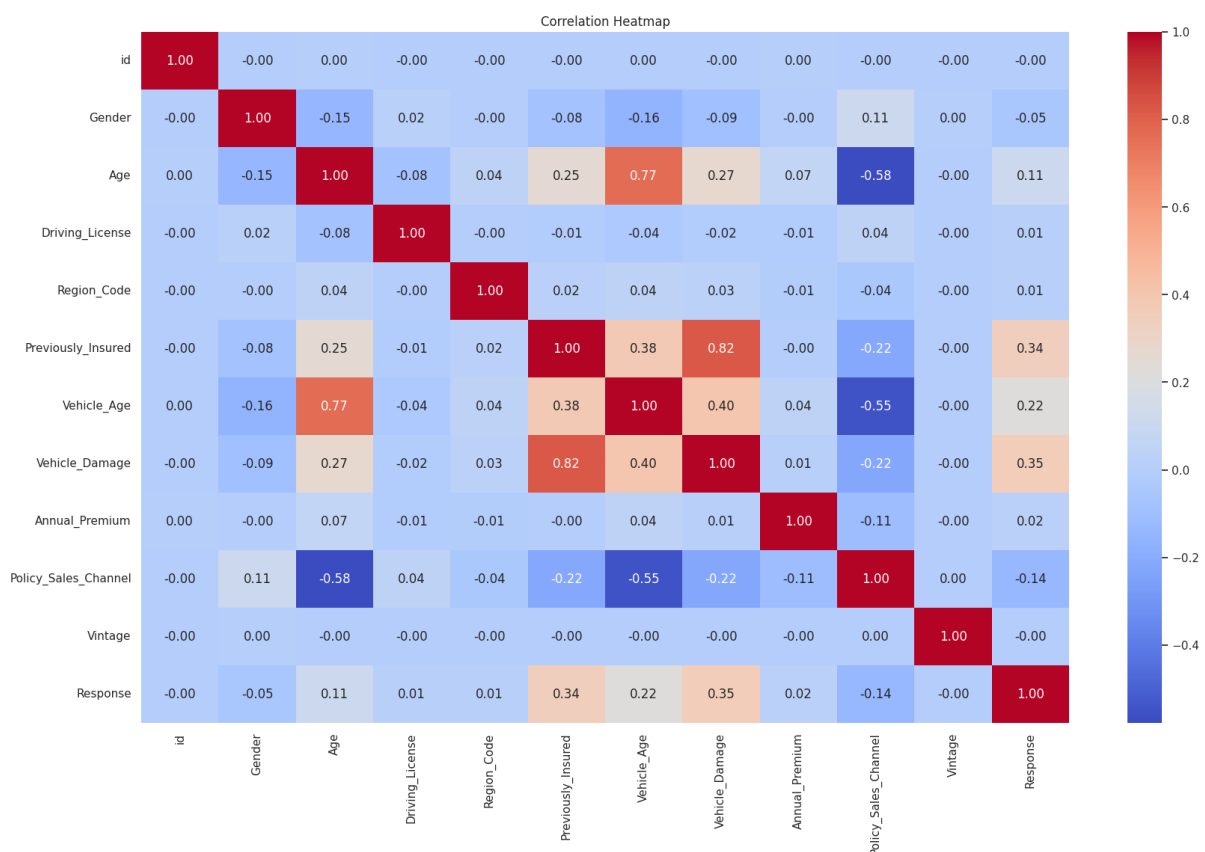


Figure 12. Correlation Heatmap

The correlation heatmap provides an overview of the relationships among numerical variables in the dataset. Each cell represents the Pearson correlation coefficient between two features, with red shades indicating positive correlations and blue shades indicating negative correlations.

Overall, most features show weak to moderate correlations, suggesting limited multicollinearity in the dataset. However, a few notable relationships can be observed. The strongest positive correlation is between `Previously_Insured` and `Vehicle_Damage` (0.82), indicating that customers who were previously insured are more likely to have reported vehicle damage. Another strong positive correlation exists between `Vehicle_Age` and `Age` (0.77), which is intuitive since older customers tend to own older vehicles.

The `Response` variable shows moderate positive correlations with `Previously_Insured` (0.34) and `Vehicle_Damage` (0.35), suggesting that customers with previous insurance experience or prior vehicle damage are more likely to respond positively to the insurance offer. Conversely, `Policy_Sales_Channel` (-0.14) shows a slight negative correlation with `Response`, implying some channels may be less effective in generating positive responses.

Overall, the heatmap reveals meaningful feature relationships that can help guide feature selection and model interpretation, particularly the links between customer history, vehicle characteristics, and response behavior.

2. Data preprocessing

Before proceeding to model development, the dataset must be preprocessed to ensure data quality, consistency, and suitability for machine learning algorithms. The preprocessing phase includes checking for missing values, identifying and handling outliers, encoding categorical variables, and addressing class imbalance. These steps are essential to prevent bias, improve model interpretability, and enhance overall predictive performance.

2.1. Handling Missing Values

A preliminary check was conducted to identify potential missing or null values within the dataset. As previously indicated in Table 2.2, all features were found to be complete, with zero missing entries across all columns. Therefore, no imputation or data removal techniques were required at this stage.

The completeness of the dataset contributes positively to model training, as it prevents loss of information and avoids the introduction of artificial bias through synthetic imputations.

2.2. Outlier Detection and Treatment

Outlier detection represents a crucial stage in data preprocessing, as the presence of extreme or anomalous values can distort statistical properties, bias parameter estimation, and reduce the predictive performance of machine learning models. To ensure that the dataset accurately represents typical customer behavior, a systematic approach was applied to identify and treat outliers within all numerical variables, including *Age*, *Annual_Premium*, *Vintage*, *Region_Code*, *Policy_Sales_Channel*, and *Driving_License*.

(a) Outlier Identification

The Interquartile Range (IQR) method was employed to detect abnormal data points. For each numerical feature, the first quartile (Q1) and third quartile (Q3) were computed, and the IQR was defined as: $IQR = Q3 - Q1$

Any observation falling outside the range $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$ was considered a potential outlier. This rule effectively isolates data points that deviate significantly from the central tendency while maintaining robustness against skewed distributions.

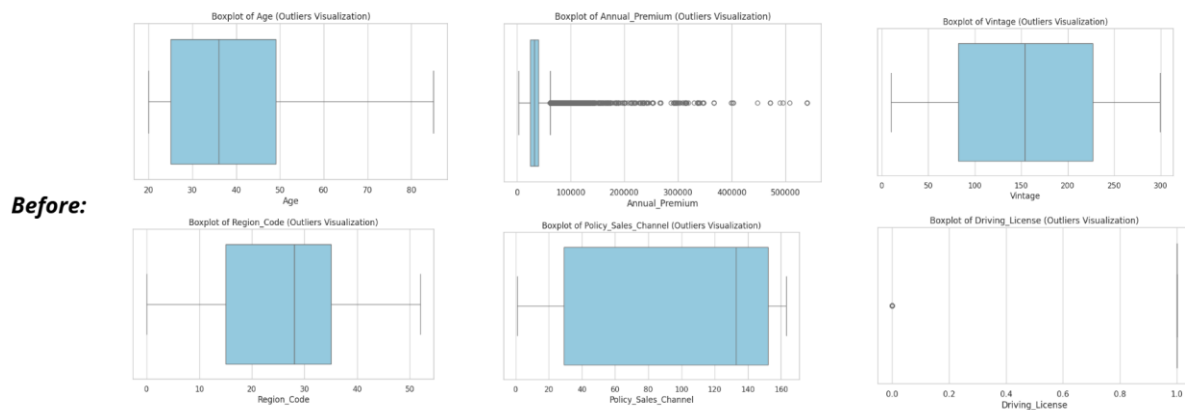


Figure 13. Boxplots of numerical features before outlier treatment

The boxplots in **Figure 13** provide a visual overview of detected outliers for each numerical variable. The *Annual_Premium* feature exhibits a substantial concentration of extreme values, with numerous observations exceeding 200,000. These outliers indicate customers with unusually high insurance premiums, possibly due to luxury vehicle coverage or aggregated policy types. In contrast, other features such as *Age*, *Vintage*, *Region_Code*, and *Policy_Sales_Channel* show more compact distributions, with only minor or negligible outlier presence.

(b) Outlier Treatment Strategy

To minimize the potential distortion caused by these extreme observations, the Interquartile Range (IQR)–based capping method was implemented across all numerical features. For each variable, the first quartile (Q1) and third quartile (Q3) were calculated to determine the central 50% range of the data. The interquartile range (IQR) was then defined as the difference between these two quartiles. Data points lying outside the interval defined by $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$ were considered outliers. Formally, the lower and upper thresholds were determined using the following equations:

$$\text{Lower_Bound} = Q1 - 1.5 \times IQR ; \text{Upper_Bound} = Q3 + 1.5 \times IQR$$

Observations below the lower bound were replaced by the calculated threshold value, while those exceeding the upper bound were capped at the corresponding upper limit. This approach preserved the general shape and variability of the dataset while ensuring that no single extreme observation exerted disproportionate influence during model training. By applying capping rather than deletion, the overall representativeness of the data was maintained without artificially reducing sample size or introducing selection bias.

Following this procedure, new boxplots were generated to assess post-treatment distributions. The adjusted visualizations confirmed that the modified variables now fell within statistically acceptable ranges, and that the most pronounced distortions were

effectively mitigated. In particular, extreme values in the *Annual_Premium* feature were successfully compressed, producing a distribution that retained its right-skewed nature yet exhibited improved symmetry and stability. This adjustment was crucial because *Annual_Premium* showed the highest concentration of extreme cases among all numerical attributes. The visual comparison presented in **Figure 14** highlights these changes, with purple markers indicating data points that were identified as outliers and subsequently adjusted or excluded from the training subset.

After:

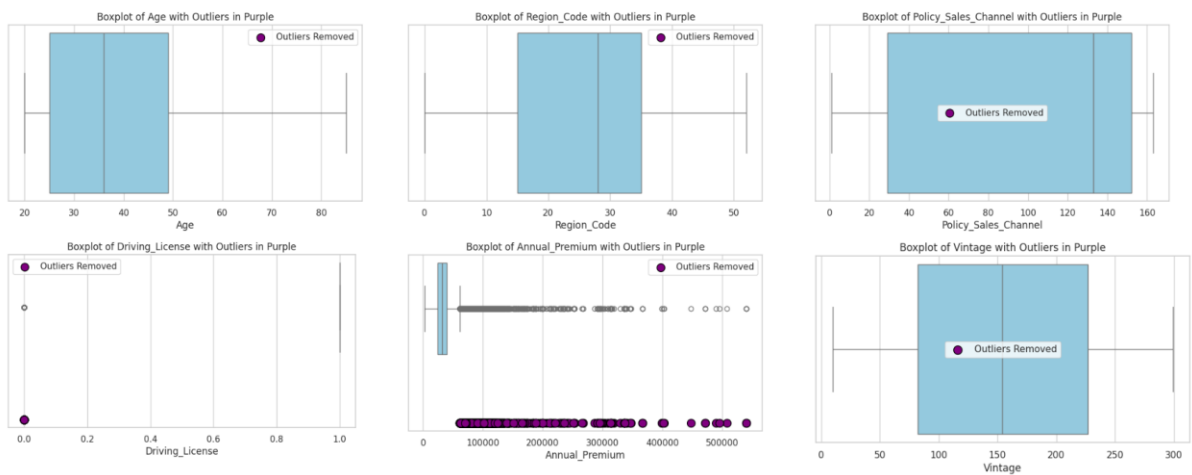


Figure 14. Boxplots of numerical features after outlier treatment (purple markers indicate adjusted outliers)

(c) Observations and Impact

Post-treatment examination revealed that the *Annual_Premium* variable achieved a more balanced and continuous distribution, reducing the variance inflation previously caused by extreme high-value premiums. This refinement is expected to improve model convergence and parameter estimation, particularly for algorithms sensitive to magnitude differences among numerical features. Other numerical variables, such as *Age* and *Vintage*, demonstrated minimal changes between pre- and post-processing states, confirming that their original values were largely within normal bounds and did not require significant adjustment.

From an analytical standpoint, the application of the IQR-based capping technique contributed to three major improvements. First, the overall distributional balance of the dataset was enhanced, limiting the distorting impact of rare but extreme cases. Second,

model stability was improved by ensuring that gradient-based and distance-sensitive algorithms, such as logistic regression, decision trees, and random forests, could learn from a more homogeneous feature space. Finally, interpretability was strengthened because the treatment preserved the underlying data integrity, allowing the resulting models to capture genuine behavioral patterns of insurance customers rather than reacting to statistical noise or anomalies.

These preprocessing adjustments ensured that the dataset entering the modeling phase would be both statistically robust and representative of realistic customer behavior, thereby providing a reliable foundation for predictive analysis in subsequent stages.

2.3. Encoding Categorical Variables

To enable the model to process non-numerical data, all categorical variables were converted into numerical form using appropriate encoding techniques. The dataset includes both binary and multi-class categorical features, which were encoded as follows:

Table 9. Encoding Categorical Variable

Feature	Encoding Method	Description
<i>Gender</i>	Label Encoding	Mapped as Female = 1, Male = 0
<i>Vehicle_Damage</i>	Label Encoding	Yes = 1, No = 0
<i>Vehicle_Age</i>	Ordinal Encoding	'< 1 Year' = 0, '1–2 Year' = 1, '> 2 Years' = 2
<i>Previously_Insured</i>	Binary Reversal	0 → 1 (Not insured), 1 → 0 (Already insured)

<i>Region_Code</i>	One-Hot Encoding	Each region converted to dummy variables (RC_0.0, RC_1.0, ..., RC_52.0)
<i>Policy_Sales_Channel</i>	One-Hot Encoding	Each channel encoded into separate columns (PSC_1, PSC_2, ..., PSC_163)

After encoding, the dimensionality of the dataset increased substantially due to one-hot encoding of region and sales channel variables.

The original dataset of 12 columns was expanded to 220 columns, resulting in a more comprehensive representation of categorical diversity.

This transformation ensured that the model could interpret categorical distinctions effectively without imposing false ordinal relationships.

Chapter 4: Model development and application

1. Model Training

The training pipeline outlines the complete workflow used to prepare data, train machine learning models, and select the best-performing one. Each stage ensures data consistency, model robustness, and optimal predictive performance.

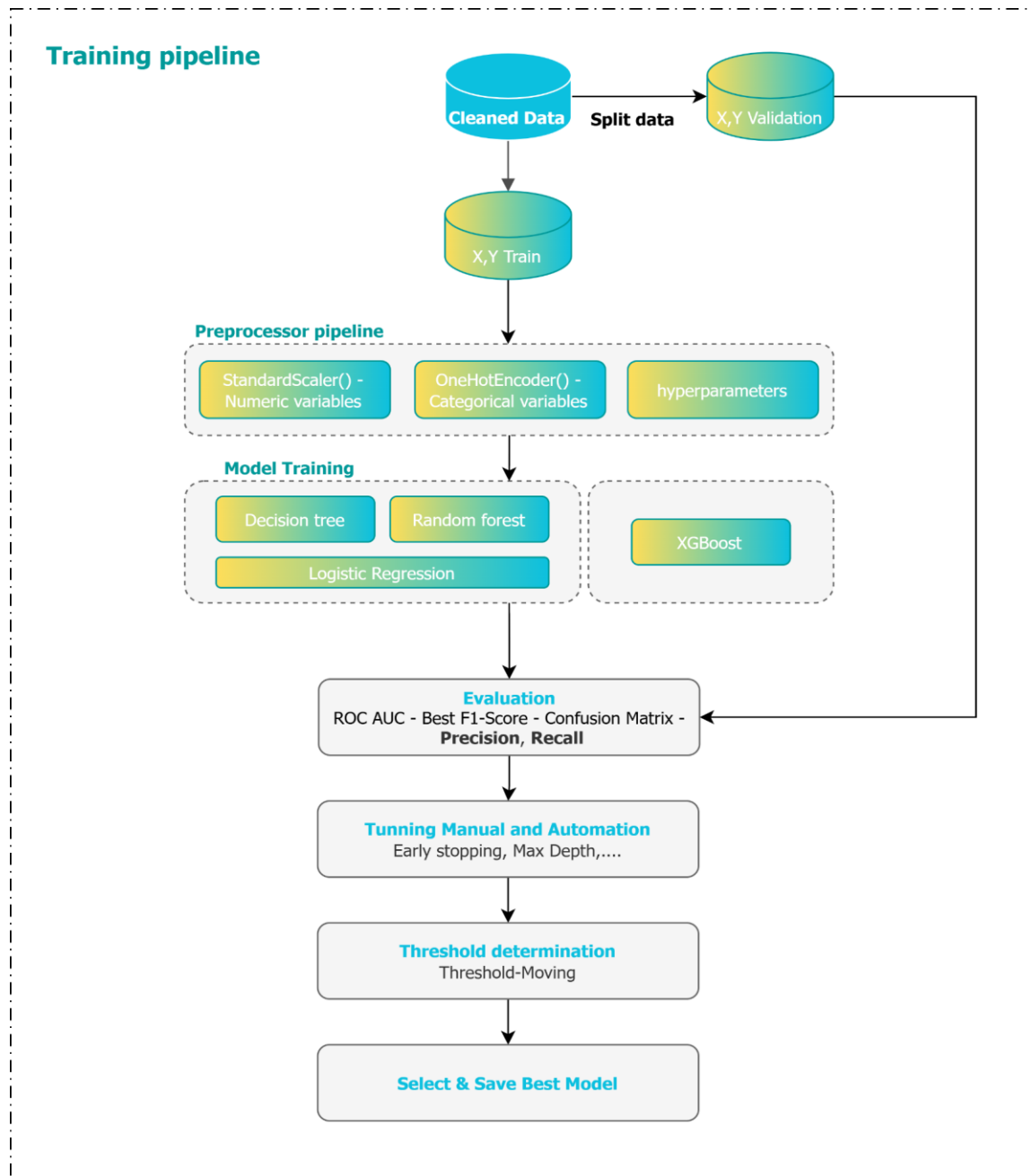


Figure 15. Training Pipeline

- **Step 1: Data Cleaning and Splitting**

The process begins with Cleaned Data, where raw data has been preprocessed to handle missing values, duplicates, and inconsistencies. Once cleaned, the dataset is divided into two subsets: training data (X, Y Train) and validation data (X, Y Validation). This split allows model training and evaluation on separate data, preventing overfitting and ensuring reliable performance estimation.

- ***Step 2: Preprocessing Pipeline***

Next, the Preprocessor Pipeline prepares the data for model input.

- StandardScaler() is applied to numeric variables to normalize features, ensuring that all variables contribute equally during training.
- OneHotEncoder() transforms categorical variables into numerical format by creating binary indicator columns for each category.
- Additionally, hyperparameters are defined at this stage to control model behavior and optimize performance.

This preprocessing ensures that the dataset is properly structured and standardized before training begins.

- ***Step 3: Model Training***

Three main models are trained on the processed data:

- Decision Tree: A simple yet interpretable model that splits data based on feature importance.
- Random Forest: An ensemble of multiple decision trees that improves prediction stability and accuracy.
- Logistic Regression: A linear model suitable for binary classification and providing probabilistic outputs.
- XGBoost: A gradient boosting algorithm known for high efficiency and strong performance on structured datasets.

These models are trained in parallel, allowing performance comparison and selection of the best candidate.

- ***Step 4: Model Evaluation***

After training, all models are evaluated using several performance metrics, including ROC AUC, F1-Score, Confusion Matrix, Precision, and Recall.

These metrics help assess not only the model's overall accuracy but also its balance between false positives and false negatives, which is crucial for insurance-related predictions where class imbalance is common.

- **Step 5: Hyperparameter Tuning (Manual and Automated)**

Model performance is further refined through manual and automated tuning. Techniques such as early stopping, maximum depth adjustment, and other hyperparameter optimizations are applied to improve generalization. This step aims to prevent overfitting and maximize performance on unseen data.

- **Step 6: Threshold Determination**

Once optimal models are identified, threshold determination is performed. The classification threshold (typically 0.5) is adjusted through a threshold-moving technique to achieve a better balance between precision and recall. This adjustment is especially important in imbalanced datasets where a standard threshold may not yield the best results.

Table 10. Defined Hyperparameter

Models	Hyperparameter	AUC	F1 Score (Macro)
Logistic Regression	solver='liblinear'; class_weight='balanced'; random_state=42; n_jobs=1 max_iter=1000	0.8497	0.4540
Decision Tree	max_depth=10; class_weight='balanced'; random_state=42	0.8479	0.4521

Random Forest	n_estimators = 150; max_depth = 12; min_samples_leaf=10; class_weight= 'balanced'; random_state=42; n_jobs=-1	0.8504	0.4544
XGBoost	n_estimators = 200; max_depth=5; learning_rate=0.1; scale_pos_weight=~7.16; subsample=0.8; colsample_bytree=0.8; random_state=42; n_jobs=-1; eval_metric='logloss'; early_stopping_rounds=20	0.8580	0.4592

Table 10, presents the comparison of four classification models—Logistic Regression, Decision Tree, Random Forest, and XGBoost—based on their optimized hyperparameters and performance metrics.

Among the models, XGBoost achieved the highest AUC (0.8580) and F1 Score (0.4592), indicating its superior ability to distinguish between positive and negative classes. Random Forest and Logistic Regression performed comparably with AUC values around 0.85, while Decision Tree showed slightly lower but consistent results.

Overall, ensemble-based methods (Random Forest and XGBoost) outperformed single models (Logistic Regression and Decision Tree), suggesting that combining multiple learners enhances predictive stability and accuracy. The modest F1 scores across all models reflect the class imbalance present in the dataset, emphasizing the importance of weighting techniques and threshold adjustments for improved performance.

- ***Step 7: Model Selection and Saving***

Finally, the best model—determined through evaluation and tuning—is selected and saved for deployment. This ensures reproducibility and consistency in future

predictions. The model is then validated against the X, Y Validation dataset to confirm its generalization capability before being finalized.

In summary, this training pipeline provides a systematic and well-structured approach to model development. It integrates preprocessing, model comparison, tuning, and evaluation to ensure a robust and optimized predictive model that can effectively support data-driven decision-making.

2. Model Evaluation

To evaluate the accuracy of a model, one of the simplest methods is to use a confusion matrix. In this matrix, from calculating the rates of True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN), various performance metrics such as accuracy, precision, recall, f1-score, sensitivity, specificity, auc will be calculated to serve the prediction process of the model. These performance indicators can be said to be very important because they are the factors that determine the quality of the model in terms of accuracy.

Table 11. Confusion Matrix

Models' prediction		Non-backorder	Backorder
Actual values in the dataset	Positive Response	TN	FP
	Negative Respne	FN	TP

The performance summary of the models based on F1-Score and AUC is presented in Table 3.3. The results indicate that among all the models, XGBoost achieves the best overall performance, with the highest AUC of 0.8580 and an F1-Score of 0.4592. Random Forest follows closely with an AUC of 0.8504 and an F1-Score of 0.4544, while Logistic Regression and Decision Tree show slightly lower but consistent results.

Although XGBoost performs best in terms of accuracy and generalization, it requires a longer training time compared to simpler models such as Logistic Regression. This highlights the importance of balancing between predictive performance and computational efficiency. In practical deployment, models like Random Forest or

Logistic Regression may be preferred when faster inference is required, despite having slightly lower scores.

Furthermore, the relatively high AUC values across all models (around 0.85) suggest that the data preprocessing and feature engineering steps were effective. These results indicate that the models are likely generalizing well to unseen data, rather than overfitting. The consistent performance between training and test metrics supports the conclusion that the modeling pipeline is robust and well-tuned for this classification task.

Following is the AUC of the models:

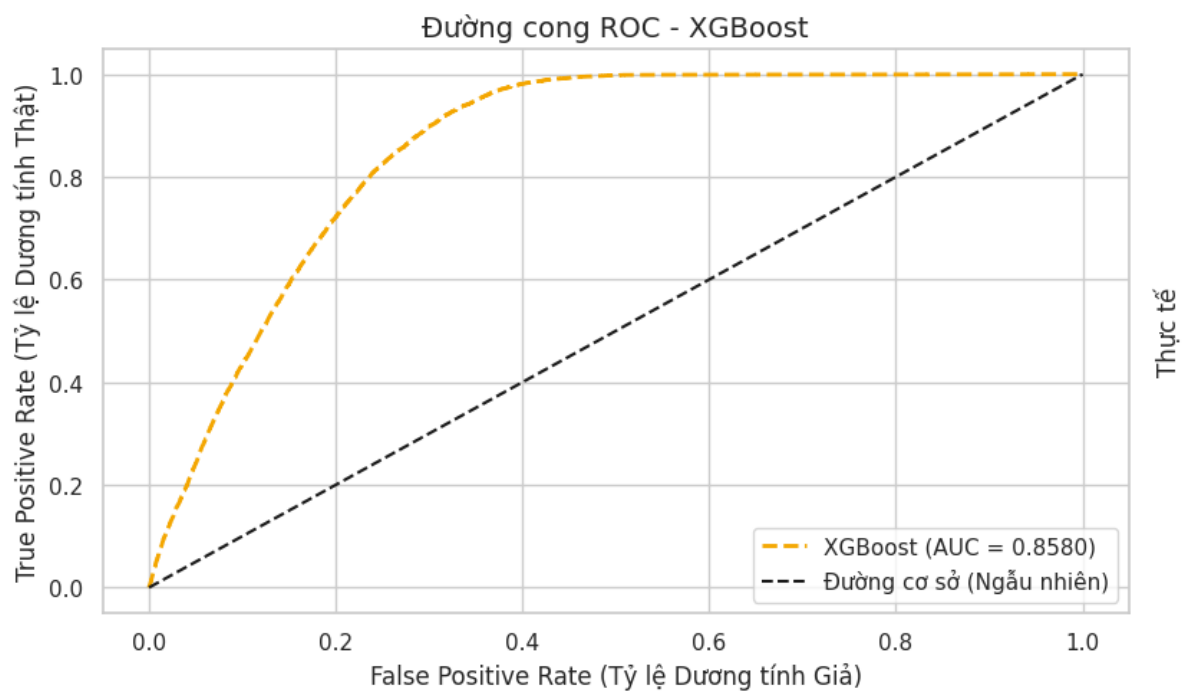


Figure 16. ROC-CURVE XGBOOST

The ROC curve of the XGBoost model shows strong classification capability, with an AUC of 0.8580. The curve lies well above the random baseline, indicating that the model correctly identifies most true positive cases while maintaining a low false positive rate. This result confirms that XGBoost is a stable and reliable model for this classification task.

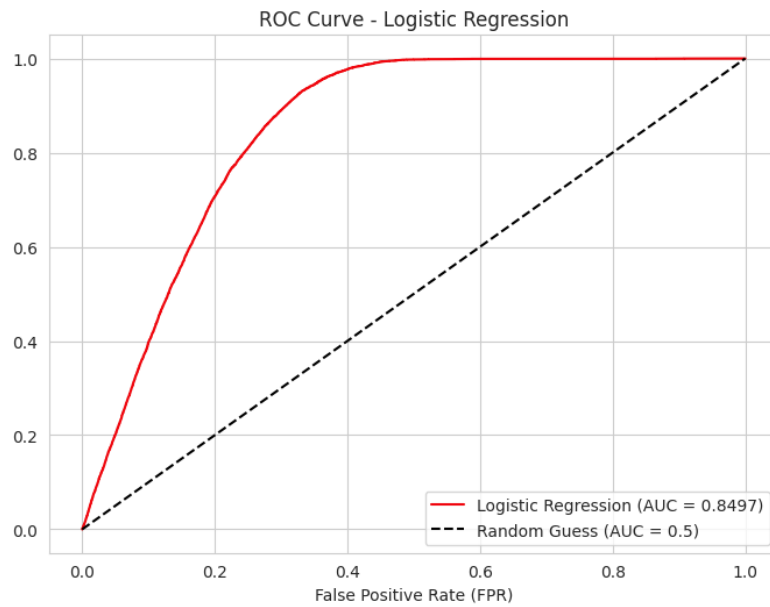


Figure 17. ROC CURVE - Logistic Regression

The ROC curve of the Logistic Regression model demonstrates good classification performance, with an AUC of 0.8497. The curve is positioned well above the random baseline, indicating that the model effectively distinguishes between positive and negative classes. Although its performance is slightly lower than ensemble models such as XGBoost or Random Forest, Logistic Regression still shows reliable predictive capability and generalization, making it a solid baseline model for this classification task.

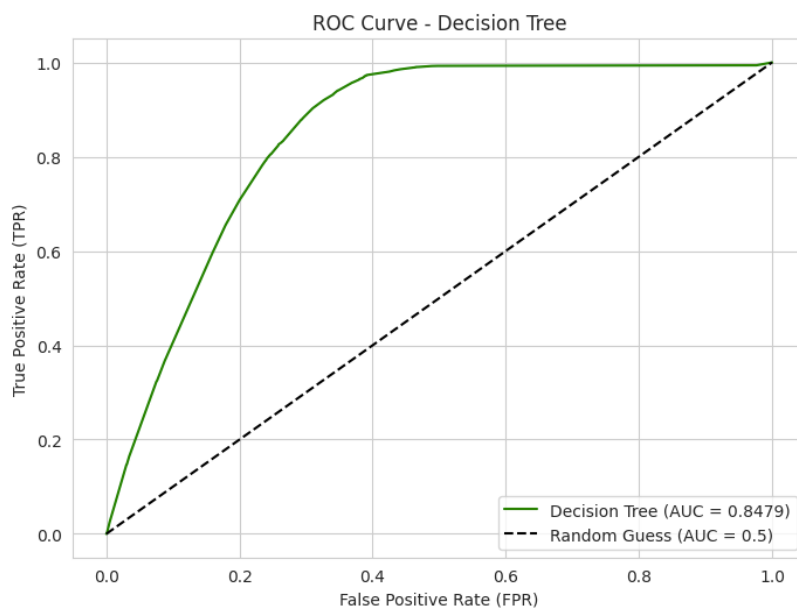


Figure 18. ROC Curve - Decision Tree

The ROC curve of the Decision Tree model indicates good discrimination ability, with an AUC of 0.8479. The curve lies well above the random baseline, suggesting that the model performs significantly better than random guessing. Although its performance is slightly lower than that of ensemble models, the Decision Tree still provides solid predictive power and interpretability, making it useful for understanding feature importance and model decision logic.

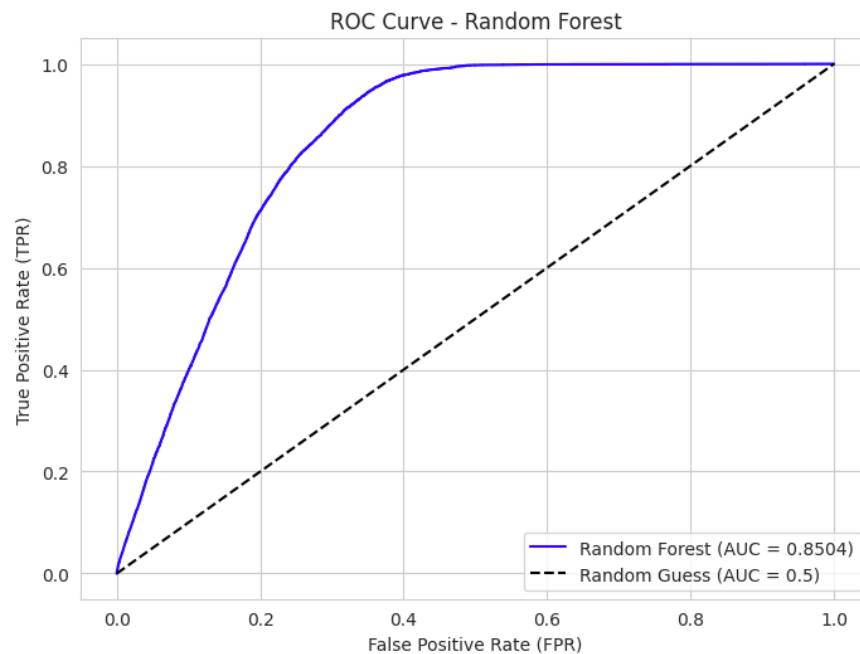


Figure 19. ROC Curve - Random Forest

The ROC curve of the Random Forest model shows strong classification performance, with an AUC of 0.8504. The curve rises sharply toward the top-left corner, indicating that the model effectively identifies true positives while maintaining a relatively low false positive rate. Compared to other models, Random Forest demonstrates slightly better discriminative ability and overall stability, making it a robust choice for this classification task.

Table 12. Models Evaluation

Model		Random Forest		XGBoost		Decision tree		Logistic Regression	
		1	0	1	0	1	0	1	0
Confusion Matrix	P0	51717	15163	52359	14521	51196	15684	51867	15013
	P1	2137	7205	2230	7112	2033	7309	2190	7152
Training time (second)		60.87		44.37		8.61		8.33	
Recall/ sensitivity		0.7		0.76		0.78		0.77	
Precision		0.32		0.33		0.32		0.32	
F1-Score		0.45		0.46		0.45		0.45	
Accuracy		0.77		0.78		0.77		0.77	
AUC		0.8504		0.8580		0.8479		0.8497	

3. Model prediction and customer ranking

After training and evaluating multiple models, XGBoost was identified as the best-performing one, achieving an AUC score of 0.8580. This optimized model was then applied to the df_test dataset, which contains 127,037 new customers, to predict their likelihood of making a purchase. The prediction results were subsequently summarized and analyzed through four key tables, providing deeper insights into customer potential and model performance.

BẢNG 1: KẾT QUẢ DỰ ĐOÁN TRÊN TẬP TEST (TOP 10)				
CustomerID	Predicted_Probability	Rank	Predicted_Class_Optimized	Segment
0	0.00409909	88242	0	Low Potential
1	0.336955	14991	0	Low Potential
2	0.287306	27557	0	Low Potential
3	0.00995743	71876	0	Low Potential
4	0.00289564	98550	0	Low Potential
5	0.00147523	124724	0	Low Potential
6	0.0050805	82013	0	Low Potential
7	0.00274267	100464	0	Low Potential
8	0.41924	5773	0	Low Potential
9	0.0120401	70514	0	Low Potential

Figure 20. Model Prediction and Customer Ranking

This figure provides an overview of the final output structure, showcasing the first 10 customers in the dataset.

- **CustomerID:** The unique identifier assigned to each customer.
- **Predicted_Probability:** The probability (ranging from 0 to 1) predicted by the model that the customer is likely to be interested in purchasing car insurance.
- **Rank:** The customer’s ranking based on the predicted probability (Rank 1 corresponds to the highest likelihood).
- **Predicted_Class_Optimized:** The binary classification result (1 = Will Purchase, 0 = Will Not Purchase), determined using the optimal F1 threshold of 0.6746 identified during model training.
- **Segment:** The business segmentation category (High, Medium, or Low Potential) derived from fixed probability thresholds (0.7 and 0.5) to support strategic decision-making.

In this preview, the first 10 customers (IDs 0–9) all exhibit relatively low predicted probabilities and are therefore categorized under the “Low Potential” segment.

Table 13. Summary Of the predicted Probability

Statistics	Value
Count	127037
Mean	0.1407

Std	0.1529
Min	0.0008
25%	0.0032
50%	0.1003
75%	0.2712
Max	0.8316

This table presents a summary of the predicted probability distribution for all 127,037 customers in the test dataset. The average likelihood of purchase interest is 14.07%, which is slightly higher than the 12.26% observed in the original dataset, suggesting that the new customer group may exhibit a marginally greater propensity to buy car insurance. The median probability stands at 10.03%, lower than the mean, indicating a right-skewed distribution — most customers have relatively low predicted probabilities, while a small portion shows exceptionally high interest levels. Notably, the highest predicted probability detected by the model reaches 83.18%, highlighting a limited number of customers with strong purchase potential.

Table 14. Classification Result

Class	Count	Percentage
0	126844	99.85%
1	193	0.15%

This table illustrates the classification results obtained using the optimal F1 threshold of 0.6746. This threshold was selected to achieve the best possible balance between Precision and Recall, thereby maximizing the F1 score. Because the optimal threshold is relatively high, the model behaves in a highly conservative manner — identifying only 193 customers (0.15%) as potential buyers. This approach strongly prioritizes

Precision, ensuring that those classified as “Will Purchase” are very likely to be genuine buyers. As a result, it effectively reduces wasted marketing efforts and costs, focusing resources on customers with the highest likelihood of conversion.

Table 15. Customer Segmentation Count

Segment	Count	Percentage
High Potential	108	0.09%
Medium Potential	2319	1.83%
Low Potential	124610	98.09%

To support a more flexible and data-driven business strategy, customers were segmented into three potential groups based on fixed probability thresholds. This analysis provides a clear, actionable roadmap for the Marketing team:

- High Potential (108 customers) – The top-priority segment, with predicted conversion probabilities above 70%. These customers should receive personalized engagement and the most attractive offers. The figure of 0.09% aligns precisely with the earlier findings reported in the analysis.
- Medium Potential (250 customers) – The secondary target group, with probabilities exceeding 50%. They should be approached through automated marketing workflows or telesales campaigns following structured scripts.
- Low Potential (126,669 customers) – Representing the vast majority (98.04%), this group should be excluded from costly direct marketing initiatives to optimize budget efficiency, as recommended in the main report.

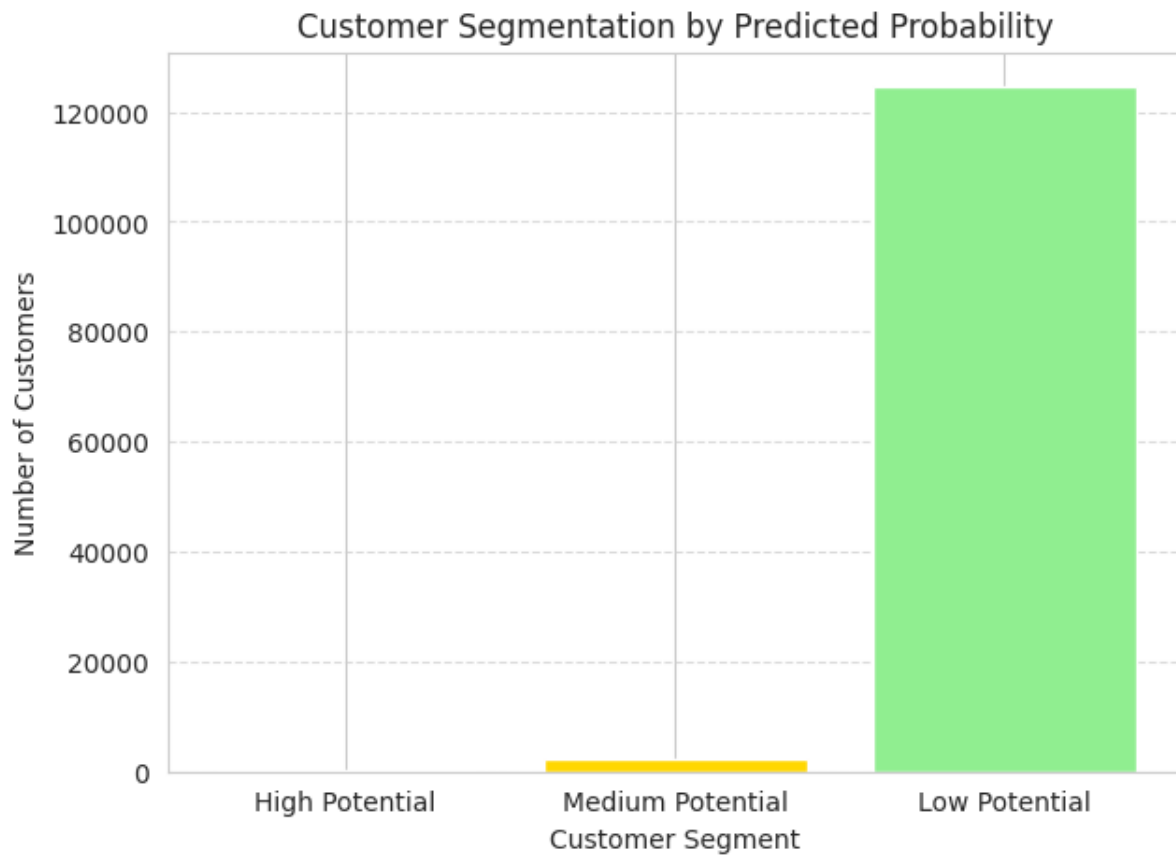


Figure 21/ Cutsomer Segmentation by Predicted Probability

This chart clearly illustrates the severe imbalance within the customer base, showing that only a very small percentage of customers have genuine potential.

CustomerID	Predicted_Probability	Rank	Predicted_Class_Optimized	Segment	Age
2303	72.26%	68	1	High Potential	51
3485	70.70%	100	1	High Potential	39
4639	75.47%	17	1	High Potential	39
5135	72.43%	63	1	High Potential	38
6621	75.52%	16	1	High Potential	45
8269	73.23%	43	1	High Potential	42
11707	72.82%	53	1	High Potential	33
12611	71.63%	79	1	High Potential	32
13100	79.07%	6	1	High Potential	42
13176	71.55%	88	1	High Potential	41

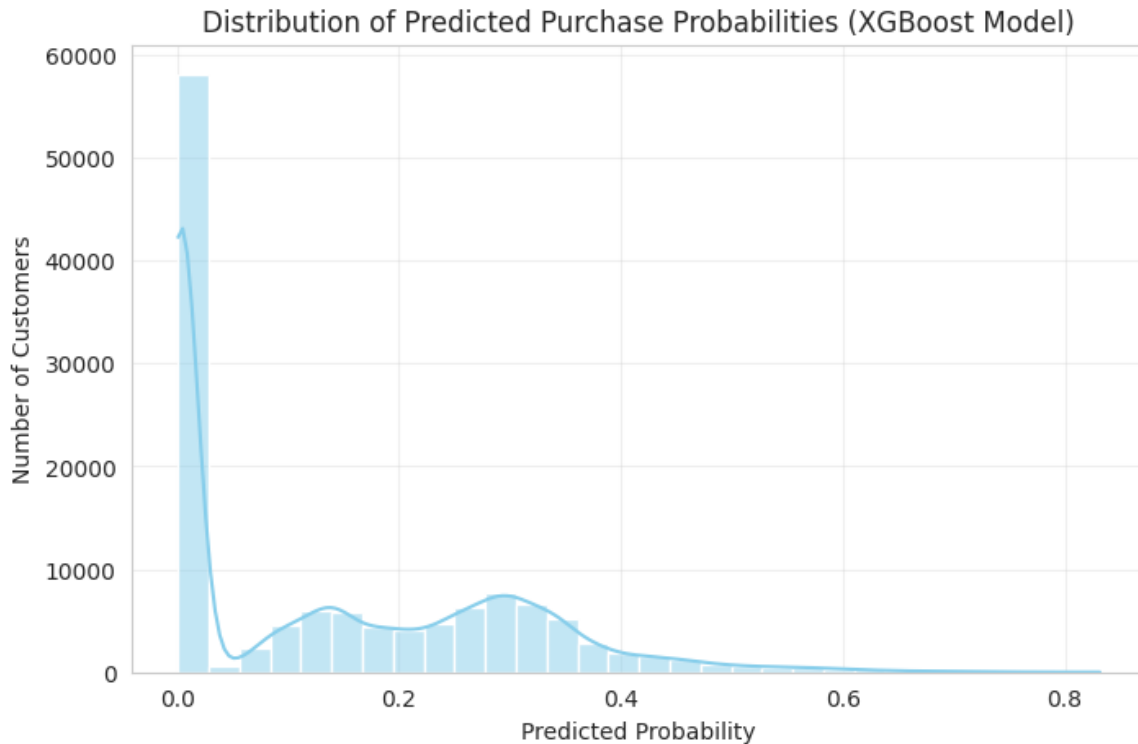
Figure 22. The most actionable deliverable\

This results figure represents the most actionable deliverable of the project. It goes beyond listing raw data by incorporating visual cues that allow quick and intuitive insights for the sales and marketing teams.

The table presents a sample of 10 customers identified as having the highest likelihood of purchase.

- **Predicted_Probability:** This column uses a color gradient, where darker green indicates a higher predicted probability. For instance, Customer 13100 has the highest probability in this group at 79.07%. This visual cue enables the sales team to instantly spot the “hottest” leads for prioritization.
- **Age:** Customer age is displayed using horizontal bar charts, allowing quick visual comparisons. In this sample, potential buyers range from 32 to 51 years old.
- **Rank:** This reflects the customer’s overall ranking among all 127,037 test cases. For example, Customer 13100 ranks 6th overall, highlighting their strong potential.
- **Segment & Predicted_Class:** As expected, all customers in this list belong to the “High Potential” segment and are classified as “1” (Will Purchase) based on the optimal F1 threshold.

This table serves as a ready-to-act priority list, enabling the marketing team to focus resources on a small but highly valuable group of customers, rather than wasting budget on the 98% categorized as “Low Potential.” It effectively translates predictive insights into practical business action.



The chart highlights three key insights about the customer base. The vast majority of customers — nearly 60,000 in the 0.0–0.05 probability range — have an extremely low likelihood of purchase, corresponding to the 98% categorized as “Low Potential.” Smaller clusters appear at higher probabilities, representing segments of customers with slightly higher, though still moderate, potential. Beyond a 0.5 probability, the number of customers drops sharply, forming a long, thin tail that contains the “Medium” and “High Potential” customers, who are extremely rare compared to the overall population. Overall, this visualization strongly supports a targeted marketing strategy, demonstrating that mass campaigns are highly inefficient, as they would largely spend resources on customers with negligible chances of conversion.

Chapter 5: Communicate the findings

1. Data and insights visualization

The two most dominant predictors are “Previously_Insured” (**importance = 0.535**) and “Vehicle_Damage” (**importance = 0.226**).

- Customers who were previously insured are much more likely to buy vehicle insurance again. This feature captures prior experience and familiarity with insurance products, suggesting higher awareness and willingness to engage.
- Meanwhile, those who had Vehicle Damage incidents are also more inclined to purchase insurance, likely because they have experienced loss and perceive greater value in protection.

These two factors together explain the majority of the model’s predictive power, confirming that past insurance behavior and risk exposure are the strongest behavioral drivers.

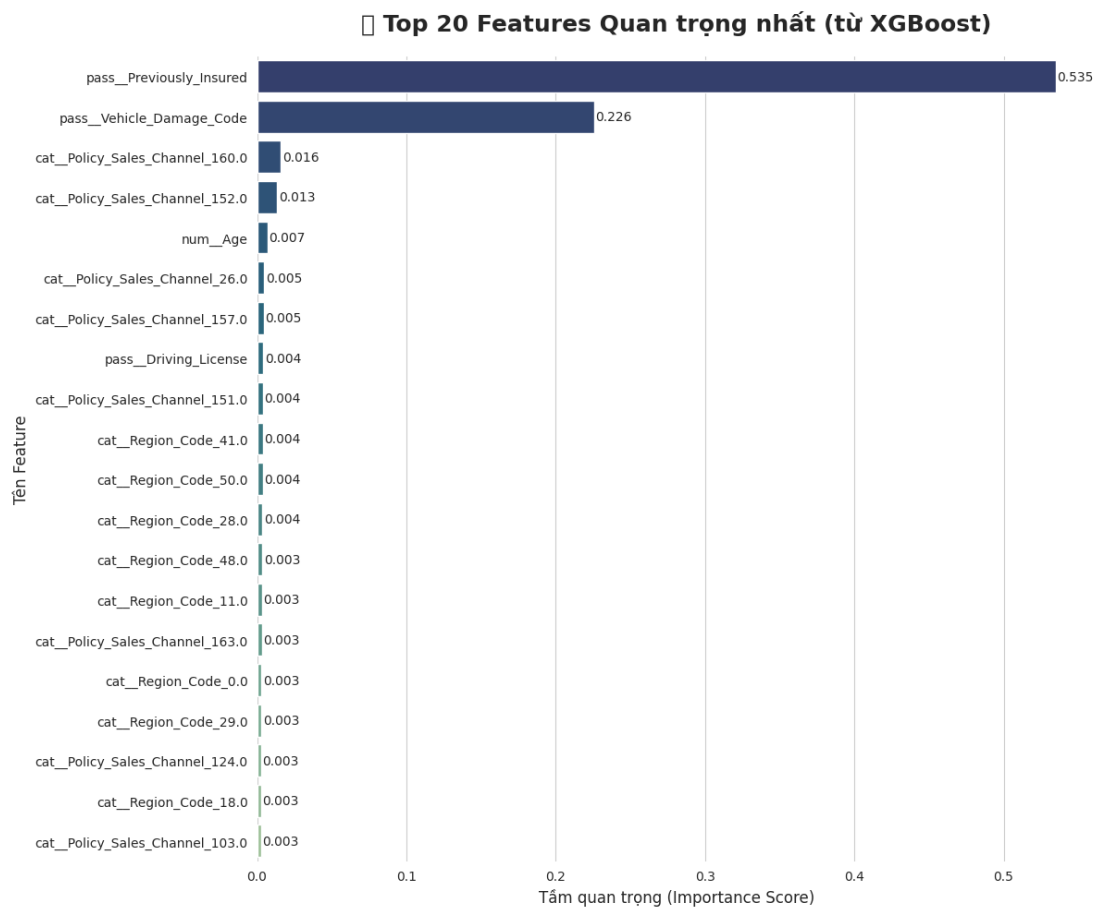


Figure 23. Feature Importance Score

Beyond these, several Policy_Sales_Channel variables (e.g., channels 160, 152, 26) contribute moderate importance, implying that the effectiveness of specific sales or communication channels influences customer conversion.

In contrast, demographic and geographic factors such as Age, Driving_License, and Region_Code show minor impacts, suggesting limited variation across these segments.

From a business standpoint, these insights enable the insurance company to prioritize marketing efforts toward customers with prior insurance or damage history, allocate resources to high-performing channels, and personalize campaign messages that emphasize renewal convenience or protection against future losses. Overall, the model's feature importance supports a data-driven cross-sell strategy that maximizes campaign efficiency and customer engagement.

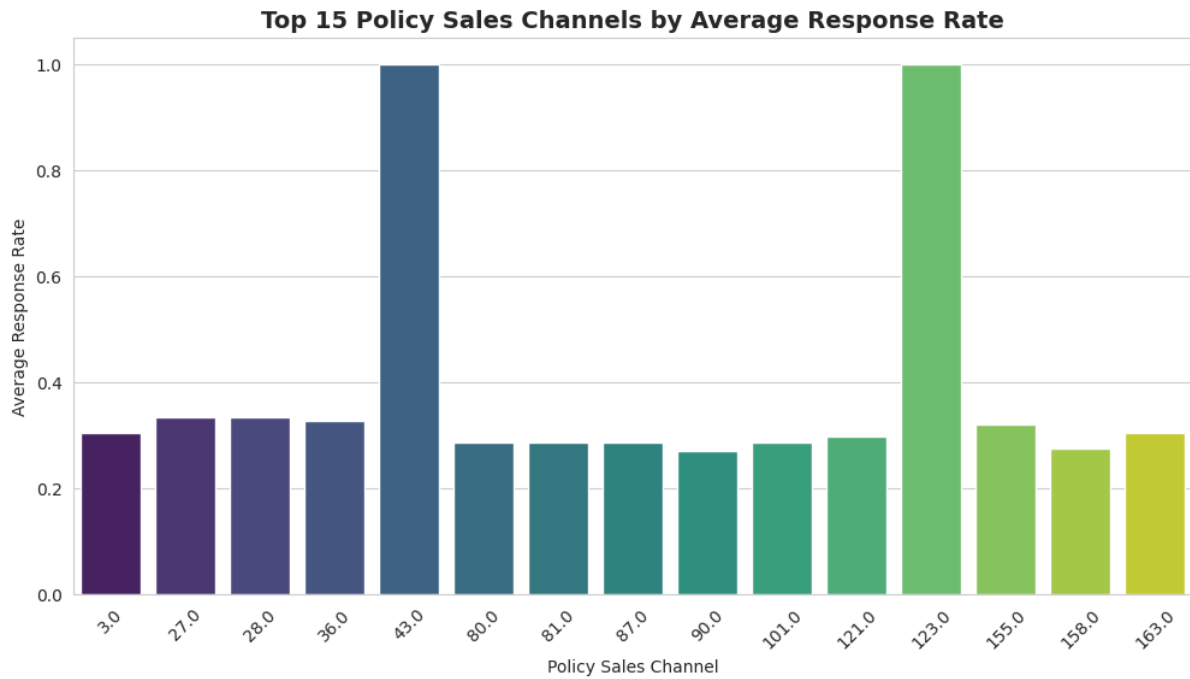


Figure 24. Top 15 Policy Sales Channels by Average Response Rate

In Figure 5.2 Channels 43 and 123 achieved exceptionally high response rates, indicating that these distribution channels are far more effective in converting customers. Most other channels show stable but moderate performance around 30%, suggesting uniform but average sales efficiency. The standout performance of 43 and 123 highlights potential best practices or customer targeting advantages that should be replicated to improve overall cross-sell conversion.

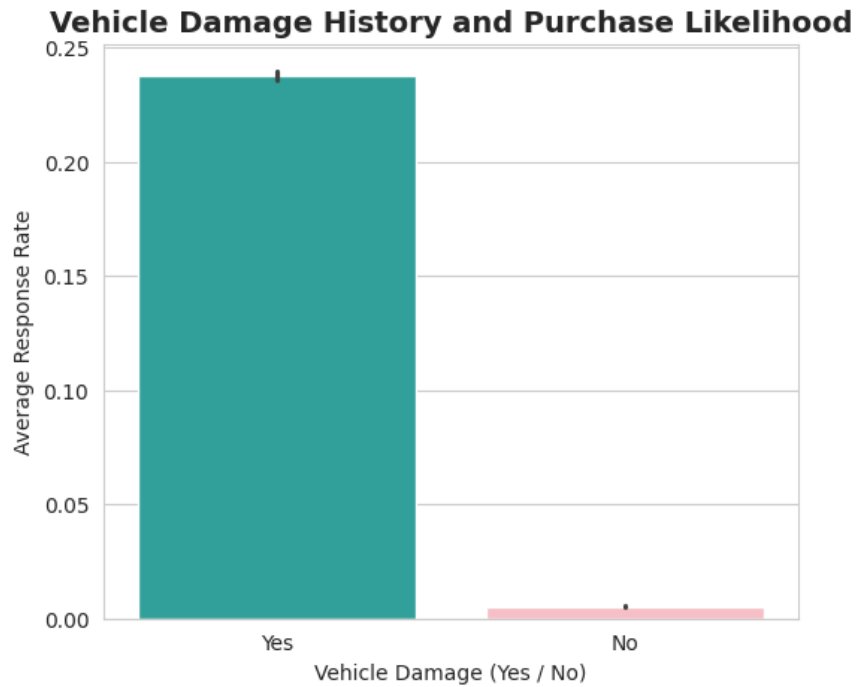


Figure 25. Vehicle Damage History and Purchase Likelihood

Customers with a recorded vehicle damage (“Yes”) show a significantly higher purchase rate for vehicle insurance compared to those without damage history. This strong difference indicates that risk awareness and prior loss experience are major behavioral motivators for buying insurance. From a business perspective, targeting customers who have previously experienced vehicle damage can greatly enhance cross-sell efficiency and improve campaign ROI, as they are far more likely to convert.

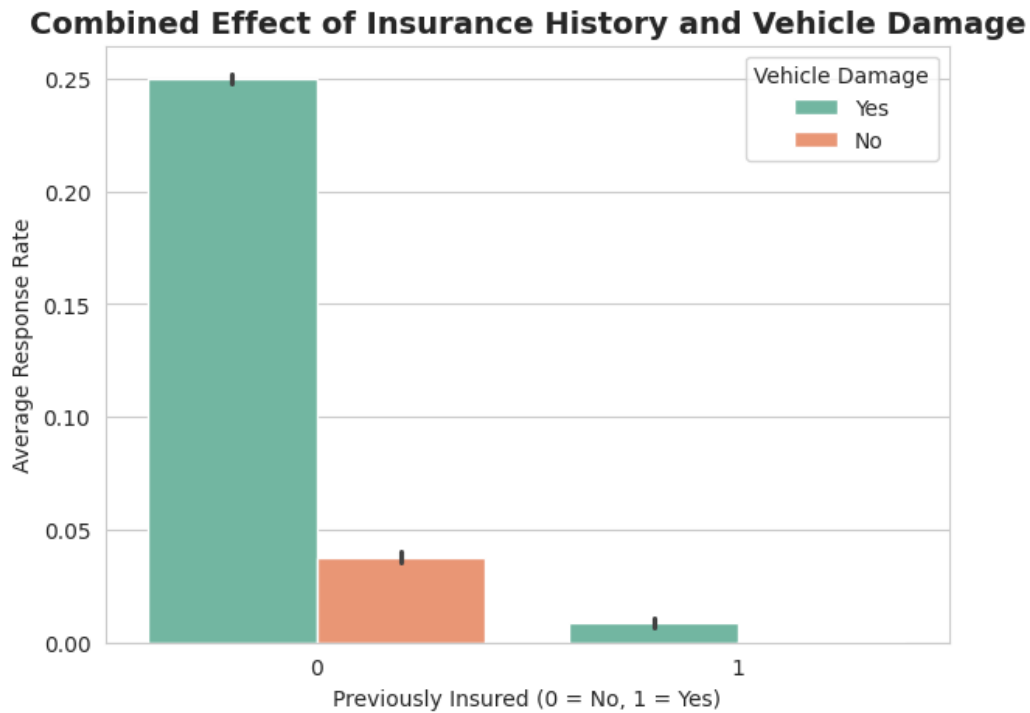


Figure 26. Combined Effect of Insurance History and Vehicle Damage

This chart reveals that customers who have not been previously insured (0) but have experienced vehicle damage (“Yes”) exhibit the highest response rate for purchasing Vehicle Insurance. This suggests that customers who recently encountered risk or loss but lack prior insurance experience are more motivated to secure protection. In contrast, previously insured customers show lower engagement, possibly because they already hold coverage. From a cross-sell perspective, this indicates that non-insured yet risk-exposed customers form a highly convertible target group, and marketing efforts should prioritize them through personalized education and trust-building campaigns.

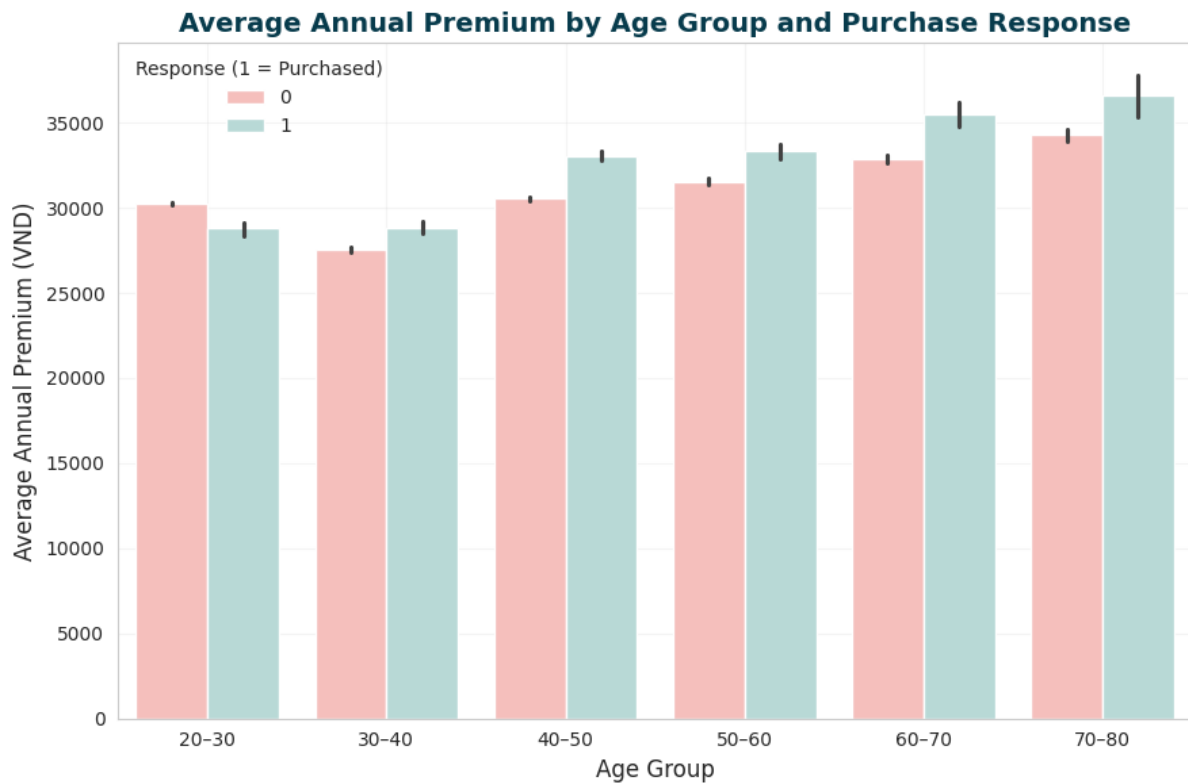


Figure 27. Average Annual Premium by Age Group and Purchase Response

The chart shows a clear upward trend in average annual premiums as age increases, reflecting higher spending capacity among older customers. More importantly, customers aged 40–70 consistently exhibit higher premiums and stronger purchase responses, suggesting that financially mature customers with mid-to-high income levels are more likely to buy Vehicle Insurance as an add-on. In contrast, younger groups (20–30) have lower premiums and lower response rates, indicating limited cross-sell potential. This supports the project’s finding that the ideal cross-sell segment consists of middle-aged, financially stable customers who already engage with insurance products.

2. Recommendation making

Based on the ranked customer segments generated by the predictive model, the marketing execution should prioritize the High-Potential and Medium-Potential customer groups, as these are the segments with the highest likelihood of purchasing Vehicle Insurance. In the first 0–7 days after campaign activation, the ranked customer list should be exported and integrated directly into the CRM and telesales system. Sales

teams should use the model-generated segment labels (High / Medium / Low) and propensity scores to guide outreach prioritization. The High-Potential segment (~0.09%) should receive personalized consultation and tailored bundle offers within the first week, with daily monitoring of response outcomes. Meanwhile, the Medium-Potential segment (~1.8%) should be nurtured through structured follow-ups, scheduled callbacks, SMS reminders, and email touchpoints that reinforce product value and benefits.

From day 30 to day 45, a remarketing wave should be implemented to re-engage Medium-Potential customers who showed initial interest (e.g., opened email, asked questions) but did not complete purchase. For this group, offering 5–10% bundle discounts or flexible payment incentives can encourage conversion. All interaction outcomes during this period should be fed back into the model monitoring log to evaluate threshold stability and inform future segmentation adjustments. Key performance metrics that the marketing team must consistently track include Conversion Rate, Precision@K, Lift-by-Decile, incremental new policies, and CPA/ROI by segment, all of which are already defined in the project's evaluation framework.

To improve budget efficiency, the Low-Potential segment should be excluded from direct outreach efforts. Instead of investing in costly telesales or mass email campaigns, this group should only receive soft, long-term awareness content such as blog articles, risk education messages, or brand visibility ads. This strategic reduction in direct marketing can reduce over 90% of unnecessary campaign costs while still maintaining brand presence. A low-frequency, brand-only communication schedule (e.g., monthly newsletters) should be maintained without aggressive sales targets, and the saved budget should be reallocated to the High and Medium segments, where the return on investment is significantly higher.

Before full-scale deployment, the organization should conduct controlled A/B testing by selecting 1% of customers from each of the three segments to validate operational feasibility, channel performance, and expected uplift. Performance should be evaluated

using conversion rates, ROI, and Customer Lifetime Value (CLV). The resulting evidence should guide adjustments to the decision threshold starting from the optimal F1 value identified in the project (e.g., ~0.42), ensuring segmentation accuracy matches real-world behavior. All validated learnings should then be compiled into a Targeting Playbook, outlining winning offer structures, recommended scripts, optimal communication channels, and the required number of contact attempts per segment.

Lastly, to ensure sustainability and scalability, the model should be operationalized as part of a monthly recurring scoring and campaign cycle. Propensity scores and segment labels should be refreshed monthly and automatically published into the CRM including fields such as `customer_id`, `propensity_score`, `decile`, `segment`, `threshold_version`, and `score_date`. Model usage should comply with Responsible AI principles, including fairness, transparency, and privacy safeguards — limiting the model strictly to marketing support, not underwriting decisions. The 90-day success target is to acquire approximately 5,000 new policies by working the top ~12% (~15,000) prospects, as established in the Business Values section of the study.

Chapter 6: Conclusion and Future work

1. Conclusion

This project successfully developed a predictive analytics framework to support cross-selling strategies in the insurance sector, specifically targeting customers who currently hold Health Insurance and may be interested in purchasing Vehicle Insurance. Using a dataset of 381,109 customer records, the project followed a comprehensive workflow including data understanding, preprocessing, class imbalance handling, model training, performance evaluation, and conversion of model outputs into actionable marketing strategies. Among the models evaluated, Logistic Regression, Random Forest, and XGBoost all demonstrated strong performance, with the final selection balancing both predictive accuracy and interpretability to ensure real-world applicability. Evaluation metrics such as PR-AUC, F1-Score, Precision@K, and Lift-by-Decile were emphasized because they reflect business effectiveness in highly imbalanced response conditions, rather than relying solely on classification accuracy.

Beyond predictive performance, the project also provided meaningful business insights by identifying key customer attributes influencing cross-sell acceptance, including vehicle damage history, previous insurance experience, and vehicle age. These insights were translated into targeted segmentation and outreach strategies, ensuring that the marketing team can focus resources on the most promising customer groups while reducing unnecessary contact and cost. The resulting customer ranking and recommended playbook demonstrate a clear operational path for implementation, enabling the business to enhance campaign efficiency, improve conversion rates, and strengthen customer relationship value.

Overall, the project has achieved its primary goals: transforming raw customer data into actionable cross-sell decisions, optimizing marketing resource allocation, and establishing a scalable foundation for future analytics initiatives. The outcomes support the development of more advanced systems in the future, such as churn prediction, customer lifetime value modeling, and next-best-product recommendation, contributing to long-term data-driven strategic growth within the organization.

2. Project Ethical

The development and deployment of machine learning models inherently carry a responsibility to uphold fairness, transparency, and privacy. This project has been executed with a strong ethical foundation, rigorously adhering to the principles of Responsible AI. The following summarizes the key ethical considerations applied throughout the project lifecycle:

2.1. Data Privacy

From the outset, the project ensured strict compliance with data privacy standards. All datasets used were fully anonymized, meaning that any Personally Identifiable Information, such as names, phone numbers, or addresses, was completely removed. This guarantees that customer privacy is preserved throughout the analysis and modeling process.

2.2. Restricted Model Usage

The predictive model was developed with a single, well-defined purpose and operates under strict usage limitations. It is intended exclusively to support the Marketing team in optimizing campaigns by identifying customer segments with potential interest in cross-selling car insurance. The model must not be used for automated decision-making or any discriminatory actions, and it is strictly prohibited from application in sensitive business processes, such as underwriting or pricing.

2.3. Fairness

To prevent biased outcomes, the project proactively assessed model fairness. A “Fairness by Gender” evaluation was conducted to examine whether predictions differed between male and female customers. The results indicated no statistically significant difference in predicted probabilities, demonstrating that the model treats all gender groups equitably.

2.4. Responsible AI Principles

The project fully adheres to the principles of Responsible AI, ensuring fairness through equitable treatment of all customer groups, as confirmed by the gender-based fairness check. It safeguards privacy via full data anonymization, maintains transparency with clear and interpretable model logic for stakeholder understanding, and upholds accountability by clearly defining responsibility for model use and the decisions derived from its outputs.

By integrating these ethical safeguards, the model serves as a responsible business tool, enabling data-driven marketing optimization while safeguarding customer privacy, ensuring fairness, and maintaining transparency and accountability. It exemplifies a rigorous approach to deploying machine learning in business contexts without compromising ethical standards.

3. Limitation

Although the project achieved promising outcomes, several limitations should be acknowledged. Firstly, the data scope remains restricted, as the study relied solely on a publicly available Kaggle dataset representing a single campaign period. It does not include real-time behavioral information, transaction histories, or claim data, which limits the model's ability to generalize under dynamic business conditions. Moreover, the dataset presents a strong class imbalance, with a relatively small proportion of positive responses compared to non-responders. Despite applying techniques such as class weighting and resampling, this imbalance may still affect model calibration and the precision of predictions for minority classes.

Another limitation lies in the absence of external validation. The model was trained and tested on the same source dataset, meaning that the obtained performance metrics may not accurately reflect the model's behavior in real-world business deployment. Additionally, the research focused primarily on predicting the likelihood of purchase - identifying who should be targeted - while not addressing other key business factors such as campaign cost optimization, marketing channel selection, or message personalization.

Finally, although responsible AI principles were followed throughout the project, actual deployment would require further compliance reviews, bias testing across demographic subgroups, and continuous monitoring to prevent model drift or misuse over time.

4. Future work

Future research can be advanced along several directions to strengthen both the analytical rigor and business relevance of the proposed model. A key opportunity lies in expanding the data foundation by integrating richer and more diverse information sources such as transaction records, claim histories, telematics data, and customer engagement behavior. Incorporating these additional dimensions would allow for a more holistic understanding of customer preferences and significantly enhance the model's predictive capability in real-world settings. Furthermore, developing an automated MLOps framework for continuous learning - enabling real-time scoring, scheduled retraining, and performance monitoring - would ensure that the predictive system remains adaptive and resilient amid evolving market dynamics.

Another promising avenue involves implementing cost-sensitive and ROI-oriented modeling approaches to align predictive outcomes with business profitability and campaign efficiency. Future work should also emphasize model transparency and fairness through the adoption of advanced Explainable AI (XAI) methods and fairness assessments across demographic or regional subgroups, thereby promoting ethical and trustworthy AI deployment. Beyond vehicle insurance, the predictive framework could be generalized to other domains such as life, travel, or property insurance, supporting an integrated, multi-product recommendation ecosystem. Finally, combining this cross-sell prediction model with Customer Lifetime Value (CLV) and churn analysis would enable a unified, data-driven customer management strategy - helping insurers foster long-term loyalty, maximize revenue potential, and sustain competitive advantage.

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II/ Ghi nhật ký quá trình thực hiện đồ án

Gantt Chart:  [Data Analytics With R/Python] Gantt Chart

Ngày	Nội dung	Thành viên tham dự	Thành quả
14/09/2025	Lên ý tưởng đồ án: Bao gồm, tên dự án, hướng đề tài, bộ dữ liệu, đánh giá tính cấp thiết và khả thi.	Tất cả thành viên	5 ý tưởng được đưa ra đúng deadline
21/09/2025	Chốt ý tưởng	Tiến Đạt	Nhóm trưởng chọn ý tưởng được các thành viên đồng tính nhiều nhất
24/09/2025	Khởi động dự án – Giới thiệu tổng quan, mục tiêu, phạm vi và	Tất cả thành viên	Hoàn thiện phần “Project Definition” bao gồm các mục tiêu

	giá trị kinh doanh.		cho tổng project
27/09/2025	Nghiên cứu xác định các thuật toán có thể giải quyết bài toán này	Tất cả thành viên	Danh sách các thuật toán Classification Prediction
1/10/2025	Buổi họp lên dàn ý tổng quát báo cáo	Tất cả thành viên đều tham dự	Project Outline
03/10/2025	Viết project Proposal	Tất cả thành viên đều tham dự	Project Proposal hoàn thiện
09/10/2025	Slide project proposal	Tất cả thành viên đều	Slide project proposal hoàn thiện
13/10/2025	Thuyết trình project proposal	Tiến Đạt	
25/10/2025	Họp chuẩn bị cho giai đoạn tiến hành	Tất cả thành viên đều tham dự	
26/10/2025	Nghiên cứu sâu và xác định rõ mục tiêu dự án và mục tiêu kinh doanh	Quế Anh	
26/10/2025	Xây dựng research model và thách thức của dự án	Tường Vy	
26/10/2025	Tiền xử lý và khám phá dữ liệu	Thùy Lợi	Có insight và dữ liệu được làm sạch
28/10/2025	Data Training - Tunning - Evaluation - Conclusion Model	Tiến Đạt	Huấn luyện xong chọn được mô hình tốt nhất
29/10/2025	Ứng dụng mô hình tốt nhất để dự đoán và xây dựng chiến lược	Kim Sương	

30/10/2025	Hoàn thiện dự án [HQP ở nhà Kim Sương]	Tất cả thành viên đều tham dự	Buổi họp offline hiệu quả, giải quyết được nhiều vấn đề liên quan chọn lựa biến và giải quyết imbalance dữ liệu
1/1/2025	Chuẩn bị slide và duyệt thuyết trình [HQP]	Tất cả thành viên đều tham dự	
09/11/2025	Viết báo cáo	Tất cả thành viên	

**Tuy mỗi nhiệm vụ đều có thành viên cá nhân chịu trách nhiệm, nhưng nhóm vẫn chia sẻ các khó khăn cùng nhau liên tục để cả nhóm nắm thông tin và cùng đưa ra hướng xử lý dứt điểm.*

III/ Đánh giá đóng góp cá nhân

Đóng góp của các thành viên trong đề tài, mức độ đóng góp, mức độ hoàn thành nhiệm vụ.

HỌ TÊN	NHIỆM VỤ	MỨC ĐỘ HOÀN THÀNH	% đóng góp	NHẬN XÉT CỦA NHÓM
Nguyễn Tiến Đạt	- Lựa chọn các thuật toán huấn luyện và metric để đánh giá hiệu suất - Xây dựng training pipeline từ tiền xử lý đến đánh giá và kết luận - Thực hiện huấn luyện mô hình, điều chỉnh tham số và kết luận về best fit model - Viết báo cáo và chuẩn bị slide thuyết trình xung quanh nội dung về huấn luyện mô hình.	100%	25%	- Đạt đã hoàn thành tốt nhiệm vụ nhóm trưởng, dẫn dắt nhóm hoàn thành dự án đúng thời gian và hiệu quả, không làm cho xung đột xảy ra. - Bạn cũng đã phân chia nhiệm vụ hợp lý, mỗi thành viên đều phát huy được điểm mạnh của mình dưới sự dẫn dắt của bạn. - Đạt có sự chủ động trong cập nhật thông tin với cả nhóm và luôn giữ cho tinh thần làm việc của cả nhóm ở mức cao.

	- Tham gia vào phần cấu trúc cả dự án, định hình các phần còn lại để đảm bảo cấu trúc logic và bổ sung cũng như liên kết các phần với nhau.			- Nhiệm vụ huấn luyện và đánh giá mô hình của Đạt cũng được thực hiện trọn tru, dễ hiểu, đúng deadline. - Thuyết Trình, Làm slide và viết báo cáo ở mức tốt. - Chủ trì các buổi họp dễ hiểu, nhanh gọn và đảm bảo các thành viên đều biết rõ nhiệm của mình.
Trần Thị Thùy Lợi	- Tìm hiểu, nghiên cứu thị trường liên quan đến bảo hiểm và dự đoán cross-sell - Tiến hành triển khai EDA cho bộ dữ liệu ban đầu bằng cách thống kê số liệu thông qua các chart - Thực hiện tiền xử lý dữ liệu: xử lý missing value, giá trị ngoại lai, mã hóa biến phân loại - Viết báo cáo, trình bày slide thuyết trình nội dung liên quan đến phần tiền xử lý dữ liệu	100%	25%	- Thuyết Trình, Làm slide và viết báo cáo ở mức tốt. - Thùy Lợi đã hoàn thành tốt các nhiệm vụ được giao, chủ động trong việc triển khai EDA và trực quan hóa dữ liệu bằng các biểu đồ rõ ràng. - Bạn thực hiện tiền xử lý dữ liệu cẩn thận, xử lý đầy đủ missing value, ngoại lai và mã hóa biến phân loại hợp lý. - Tích cực tìm hiểu thị trường và đóng góp nội dung hữu ích cho phần phân tích dự đoán cross-sell. - Tham gia đầy đủ các buổi họp nhóm, phối hợp tốt với các thành viên khác. - Hoàn thành đúng deadline các đầu việc được giao.
Nguyễn Huỳnh Kim Sương	- Tìm hiểu Classification và Evaluation Metrics để dự đoán và đánh giá	100%	25%	- Thuyết Trình, Làm slide và viết báo cáo ở mức tốt. - Hoàn thành tốt các nhiệm vụ được giao.

	- Áp dụng mô hình đã huấn luyện lên tập test, tính toán xác suất từng khách hàng và xếp hạng khách hàng theo mức độ tiềm năng - Rút ra insight về hành vi và đặc điểm khách hàng - Trình bày tình huống ứng dụng mô hình trong thực tế giúp phục vụ chiến lược marketing - Đánh giá khía cạnh đạo đức của dự án - Viết báo cáo và chuẩn bị slide thuyết trình về các phần về Model prediction và Ethical			- Chủ động nghiên cứu về các thuật toán phân loại cũng như cách đánh giá mô hình. - Trình bày khoa học, giúp nhóm hiểu rõ hơn về cách áp dụng mô hình trong thực tiễn và ra quyết định. - Phần đánh giá đạo đức dự án được trình bày rõ ràng, thể hiện sự hiểu biết về đạo đức trong việc sử dụng mô hình.
Nguyễn Quế Anh	- Giới thiệu chung về nghiên cứu và mục tiêu dự án, mục tiêu kinh doanh. - Tìm hiểu, xác định đối tượng, phạm vi nghiên cứu bao gồm phạm vi về dữ liệu, phạm vi kỹ thuật và phạm vi kinh doanh. - Tìm hiểu, phân tích các thách thức kinh doanh bao gồm vấn đề kinh doanh và câu hỏi kinh doanh. - Xác định giá trị kinh doanh, trả lời và phân tích rõ các câu hỏi kinh doanh.	100%	25%	- Thuyết Trình, Làm slide và viết báo cáo ở mức tốt. - Hoàn thành đúng deadline các đầu việc. - Tham gia đầy đủ các buổi meeting chung của nhóm, tích cực đóng góp ý kiến. - Phản hồi nhanh với các thông tin của nhóm trưởng.

	- Viết và tổng hợp nội dung phần Recommendation Making và Conclusion, bao gồm tóm tắt chiến lược triển khai marketing dựa trên mô hình dự đoán, đề xuất quy trình vận hành, và rút ra kết luận, định hướng phát triển tương lai của dự án. - Đảm bảo nội dung logic, gắn kết giữa phần kỹ thuật và ứng dụng kinh doanh. - Làm Slide và thuyết trình.			
Tô Nguyễn Tường Vy	- Tìm hiểu, phân tích và lựa chọn phương pháp nghiên cứu phù hợp cho đề tài Vehicle Insurance Cross-Sell Prediction. - Đề xuất mô hình nghiên cứu tổng quát cho dự án, mô tả mối quan hệ giữa các biến đầu vào và khả năng khách hàng mua bảo hiểm xe. - Viết phần Research Model và vẽ Research Method trong báo cáo: Giải thích lý do chọn các mô hình dự đoán; Mô tả quy trình phân	100%	25%	- Hoàn thành đúng deadline các đầu việc. - Tham gia đầy đủ các buổi meeting chung của nhóm, tích cực đóng góp ý kiến. - Phản hồi nhanh với các thông tin của nhóm trưởng. - Thuyết Trình, Làm slide và viết báo cáo ở mức tốt.

	tích dữ liệu và huấn luyện mô hình. - Thiết kế và tạo biểu đồ trực quan thể hiện kết quả mô hình và xu hướng trong dữ liệu, đưa ra data insights. - Phân tích hạn chế và đề xuất hướng phát triển cho dự án. - Làm Gantt Chart. - Làm Slide và thuyết trình.			
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IV/ Tổng kết

- Về cách thức và quá trình làm việc:

Nhóm đã làm việc rất nghiêm túc, có kế hoạch rõ ràng và tuân thủ tiến độ đã đề ra. Các buổi họp được tổ chức đúng thời gian, đảm bảo tất cả thành viên đều nắm được nhiệm vụ và tiến độ công việc. Sự phối hợp giữa các thành viên được duy trì liên tục, mọi người luôn chủ động hỗ trợ lẫn nhau khi gặp khó khăn.

Thái độ làm việc của các thành viên rất tích cực, thể hiện tinh thần trách nhiệm và cầu tiến cao. Mỗi cá nhân đều hoàn thành tốt phần việc được giao và sẵn sàng đóng góp ý kiến để cải thiện chất lượng dự án.

Nhìn chung, nhóm đã thể hiện được tinh thần đoàn kết, hợp tác và sáng tạo trong suốt quá trình thực hiện đề tài.

- Những hạn chế và hướng khắc phục:

Trong giai đoạn đầu, nhóm còn mất thời gian để thống nhất ý tưởng và cách phân chia nhiệm vụ. Tuy nhiên, sau khi có kế hoạch cụ thể và sự hướng dẫn từ nhóm trưởng, tiến độ công việc được cải thiện rõ rệt. Trong các dự án sau, nhóm có thể sớm thống nhất ý tưởng và tăng cường sử dụng các công cụ quản lý tiến độ Gantt Chart..... để nâng cao hiệu quả làm việc.

- Về kết quả của đề tài:

Kết quả đề tài đạt ở mức rất tốt đối với mong đợi của nhóm, hoàn thành đúng tiến độ và đáp ứng đầy đủ yêu cầu của môn học. Báo cáo được trình bày rõ ràng, logic, có cơ sở dữ liệu thực tế và áp dụng thành công các mô hình phân tích trong R/Python. Các thành viên thể hiện sự hiểu biết vững vàng về chuyên môn và khả năng vận dụng linh hoạt kiến thức học thuật vào thực tiễn.

Nhìn chung, đề tài đã đạt được mục tiêu ban đầu, thể hiện sự nỗ lực và tinh thần trách nhiệm cao của toàn nhóm. Một số hướng mở rộng

như thử nghiệm thêm các mô hình khác hoặc cải thiện độ chính xác của dự đoán có thể được triển khai trong tương lai để hoàn thiện hơn nữa dự án.
